Introduction to Chess Strategy

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Introduction

This book aims to give a brief introduction to chess strategy. Strategy is the part of chess that is concerned with long-term goals and the evaluation of positions. It is a distinct subject from tactics, which focuses on short-term tricks and forcing moves. Strategy tells us what goals we should have, while tactics help us to achieve those goals.

This book breaks strategy down into four elements: material, pawn structure, king safety, and piece activity. This division is somewhat arbitrary and you will probably encounter alternative approaches in other places. For example, another book might include initiative as an independent element of strategy, whereas I consider this to be part of piece activity. The exact approach to breaking down strategy into a number of elements doesn't matter much, just as long as it helps us start thinking about the most important strategic issues.

We can also divide chess games into three different phases: the opening comes first, the middlegame next, and the endgame last of all. The strategy of middlegames is really just the general strategy of chess and is covered in the initial chapters of this book. Opening strategy and endgame strategy have some unique qualities, so these each get separate chapters.

This book was written in response to what I see as the problems with the most commonly recommended strategy books. I have tried to keep the writing simple and straightforward, and to keep the book short. I have also tried to make it possible to read this book without needing to use a board to play out the moves. A few of the examples may be too complicated for this, but in these cases the important point should be obvious even if the use of a board isn't possible.

Of course, you won't develop a deep understanding of strategy just from reading this book. The goal of this book is to give a quick, big-picture view of strategy, which is a good first step toward future learning. To seriously improve your understanding of strategy in the long-run, you will need to play many games, learn from those games, and spend a lot of time studying the games of stronger players. However, this book should give you a good foundation of strategic ideas so that you will progress more quickly as you engage in these other forms of study.

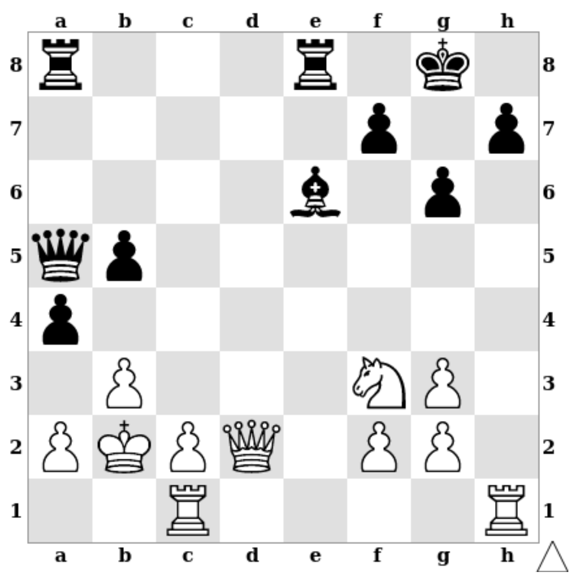
This book concludes with an appendix featuring some complete annotated games to reinforce the ideas in the main part of the book. The games are also available in a Lichess study.

1 – Choosing a Plan

Before we begin looking at the individual strategic elements of chess, we should briefly consider how to form a plan. The most important point to understand about planning is that you should choose a plan based on the nature of the position you are playing. Different positions need to be played differently, and the only way to determine the best plan for a given position is to think about that position's characteristics.

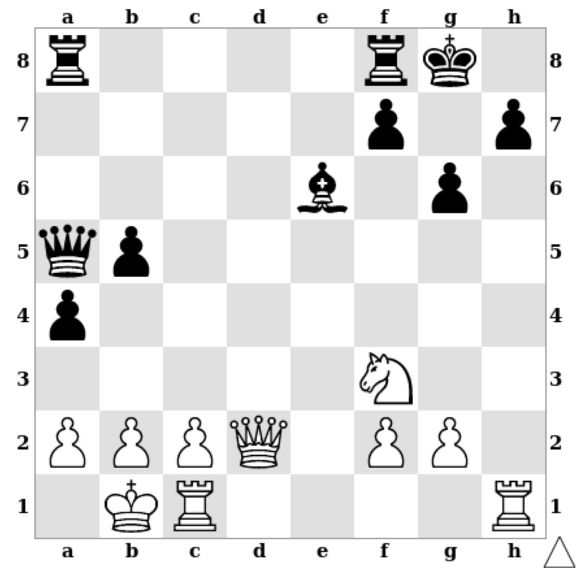
This is such a basic point that many people realize it on their own without ever being told. To these people, it will seem unnecessary to make the observation. However, there are other players who play and study chess for quite a while without fully realizing that each position has its own demands. These players have certain plans that they use over and over again, regardless of the position they find themselves in. It may be that they simply enjoy using these plans or it may be that they have never learned how to do anything else. In either case, this will often turn out to be a major obstacle toward improvement for the player in question.

One common version of this problem is when a player will try to attack the opponent's king in every single position. Such a player will do this whether he is ahead or behind in material, whether his opponent's king is weak or well-defended, etc. Having realized that trading queens makes it harder to attack in this way, he will begin avoiding queen trades as well. Obviously, this leads to his choosing the wrong plan quite often.



In this position, for instance, White has a winning advantage after 1.Qxa5 due to his extra pawn and his active pieces. If he instead avoids the queen trade and plays for an attack against Black's king with 1.Qh6?, then he is lucky to be able to draw after 1...axb3 2.Qxh7+ Kf8 3.Qh6+ Ke7 4.Qh4+ Kf8 etc. White doesn't have time for moves like 3.Ng5? and 3.Nd4? because of 3...Qa3+ 4.Kc3 Qc5+ etc. with a winning attack for Black.

Change a few details of the position, however, and the best plan changes. In this position, White has no extra pawn, but Black also has no attack of his own yet. Now 1.Qxa5? is only equal, and White should instead play 1.Qh6, which gives him a winning attack.



In both positions, White faced a choice between trading queens in order to go into an endgame and avoiding the queen trade in order to attack. In both cases, the correct choice depended entirely on the specific details of the position. Being biased toward a certain kind of plan only makes these sorts of decisions harder by distracting you from what is actually going on in the position. In other words, you can learn to choose better plans by focusing less on what you want to do and more on what the position you are playing wants you to do.

Over the further chapters of this book we will learn about a number of strategic ideas that can help in determining the best plan in a given position. As you continue to play and study chess, you will further improve your ability to find the best plan in different kinds of positions. However, the very realization that your choice of plan should come from the characteristics of the position itself, and not just from your own preferences, will already help you to start playing better chess.

2 – Material

Some chess pieces are stronger than others. Having stronger pieces is called a *material advantage*. The most common way to measure material is with these values:

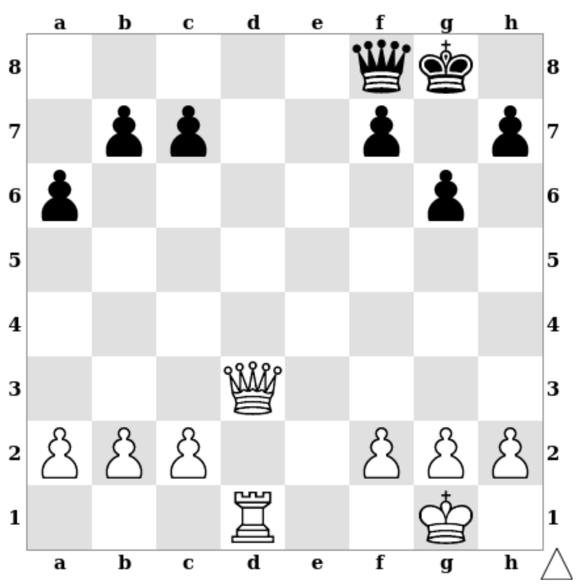
Pawn	1
Knight	3
Bishop	3
Rook	5
Queen	9

(The king isn't given a value since it can't be traded.)

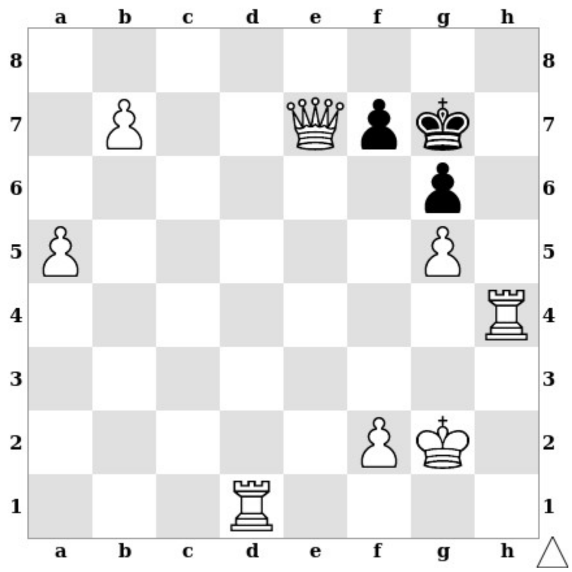
These values aren't part of the rules, they are just a way to think about the strength of each player's pieces and to think about possible trades. For example, trading a *minor piece* (a bishop or knight) for an opponent's rook is a good idea since you *win material* (about two pawns' worth).

Using a Material Advantage

As long as nothing important changes, the side with a material advantage will often win in the endgame. For this reason, when you have a material advantage you should keep the position simple and trade pieces. When your opponent has a material advantage, you should make things complicated and avoid trading pieces.

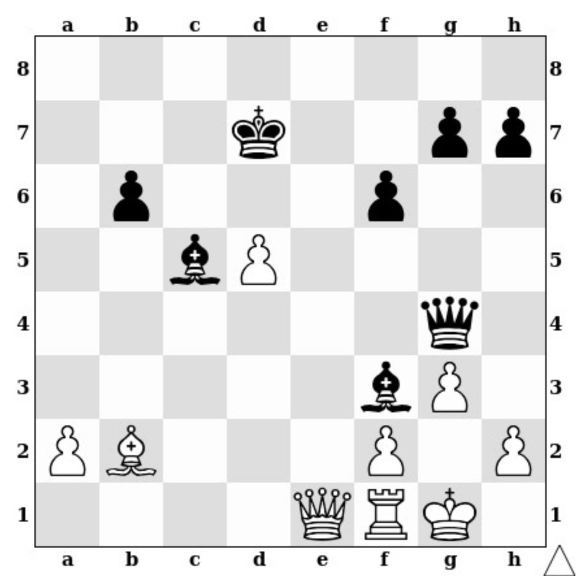


In this position, White is up five points (since he has an extra rook). He can win in many ways, but one very easy way is to play 1.Qd8. This forces a trade of queens, removing Black's only dangerous piece and leaving White with any easy win.



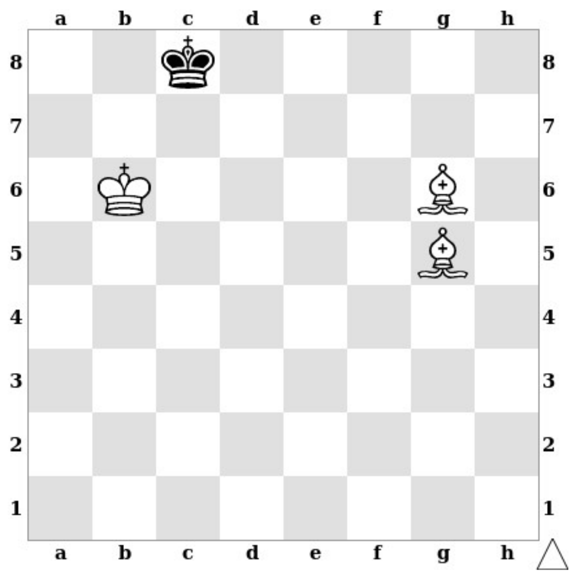
When playing with a huge material advantage, there comes a point at which it is better to use your extra pieces to force checkmate, rather than to promote more and more pawns. In this position, for example, it would be a mistake to play 1.b8=Q?, ending the game in an immediate stalemate. Instead, with such a large material advantage, White should be looking for a way to force checkmate, for example with 1.Qf6+ and 2.Rh8#. White could also play 1.Rd7 and 2.Qxf7#, though he should be sure to check that 1.Rd7 is not stalemate. In a position like this, moves that put the opponent's king in check are often safest since they will not create stalemate.

In some cases the best way to use a material advantage is to give back some of the material. Here White has three more points of material than Black. However, Black has some dangerous threats in the position, especially the threat of playing ...Qh3 and ...Qg2#. The easiest way for White to win is to give back a pawn with 1.Qe6+ Qxe6 2.dxe6 Kxe6. This eliminates Black's threats and leaves White with a two point advantage.



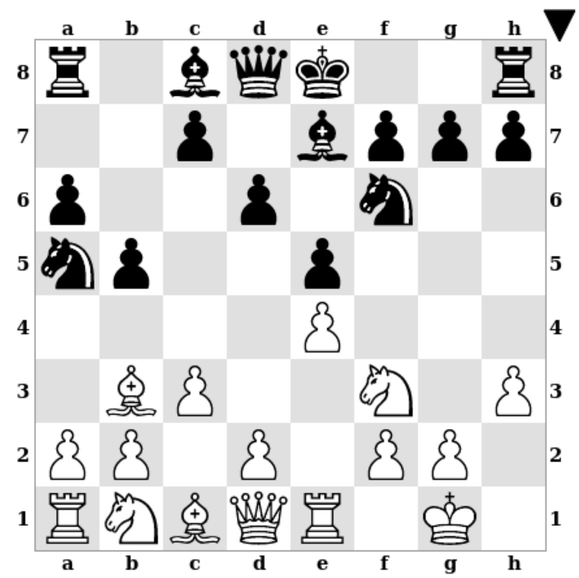
The Bishop Pair

The biggest weakness of a bishop is that it can only control squares of one color. However, when you have both of your bishops, they can work together to control squares of both colors. For this reason, having both bishops is considered an advantage, called having *the bishop pair*. Having the bishop pair is worth an extra half of a pawn on top of the value of the bishops themselves. So while a single bishop is worth three pawns, both bishops together are worth six and a half pawns. When both players have the bishop pair it isn't very interesting since the advantages cancel out. For this reason, the bishop pair is usually only mentioned when one player has it and the other doesn't.



One simple example of the power of two bishops is the ease with which they can checkmate a king on an empty board. Here White mates in three after 1.Bf5+ Kb8 2.Bf4+ Ka8 3.Be4#. Other combinations of two minor pieces either can't force mate against a lone king or can do so only with much greater difficulty.

The bishop pair is also an advantage at earlier stages of the game. In this position Black should play 1...Nxb3, trading off one of White's bishops and giving himself the advantage of the bishop pair. If White moved first, he would play 1.Bc2 to avoid this.



3 – Pawn Structure

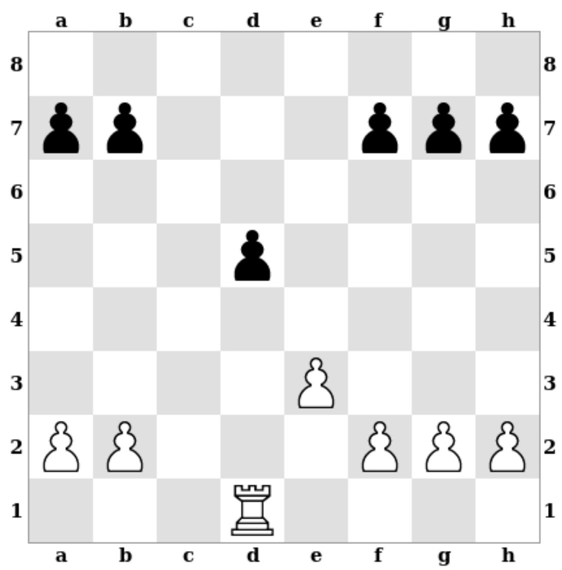
Because pawns are slow and have limited movement, they get stuck in formations called *pawn structures*. Pawn structures are important for a few reasons, including:

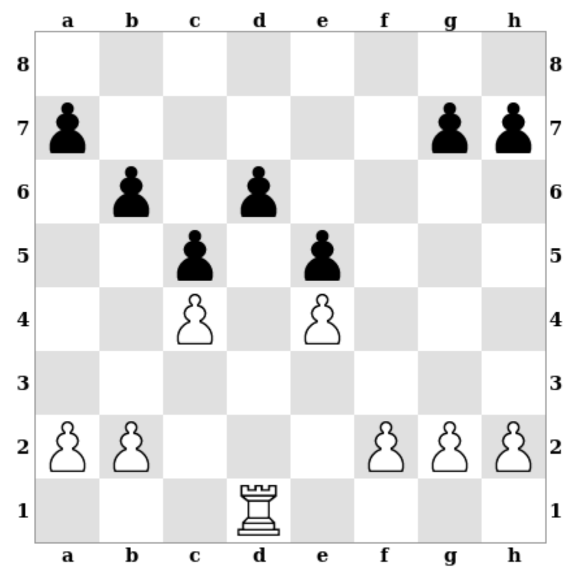
1) pawn structures can make pawns weak
2) pawn structures can make squares weak
3) pawn structures can affect the activity of other pieces
4) pawn structures can affect pawn promotion

A pawn structure may have both positive and negative effects for the same player. For example, a player's pawns may be weak, but the pawn structure may leave his other pieces very active.

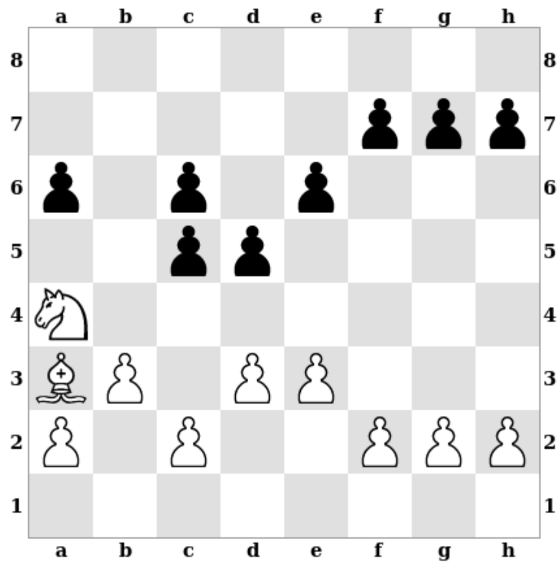
Weak Pawns

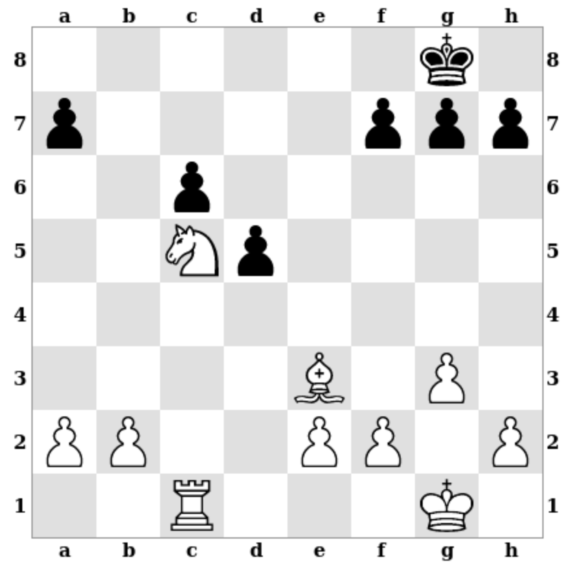
Pawns sometimes become long-term weaknesses, especially when they cannot be guarded by other pawns. There are a number of different kinds of pawns that are potential weaknesses. For example:





In the first diagram, Black's d5 pawn is *isolated* (there are no black pawns on the adjacent files). In the second diagram, Black's d6 pawn is *backward* (the black pawns on adjacent files have moved past it).

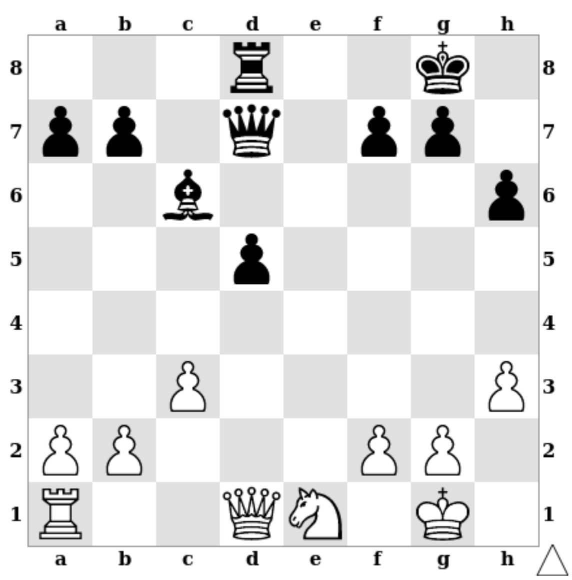




In the first of these positions, Black's pawns on c5 and c6 are *doubled pawns* (two pawns of the same color on the same file). In the second position, Black's pawns on c6 and d5 are *hanging pawns* or an *isolated pawn pair* (a pair of pawns that are isolated together).

Pawns are not necessarily serious weaknesses just because they fall into one of these categories. The actual weakness of pawns is determined by how difficult they are to defend and how easy they are to attack. For example, a pawn weakness is often more of a problem when the opponent has an open file leading to it since this makes it easier for him to attack it.

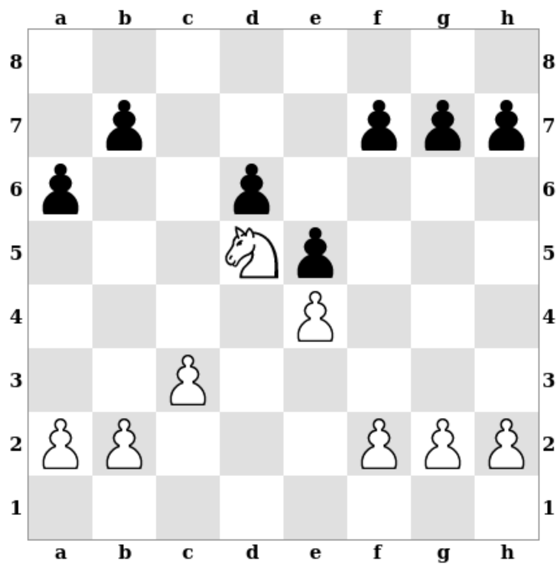
Playing Against Weak Pawns



When playing against a weak pawn, you should control the square in front of the pawn to stop it from advancing. In this position, Black has an isolated pawn on d5 that is both weak and is making his bishop rather passive. If Black moves first he should play 1...d4, eliminating his weak pawn and activating his bishop. If White moves first instead, then he should play 1.Nf3 or 1.Nc2 to control the d4 square and stop Black from playing ...d4.

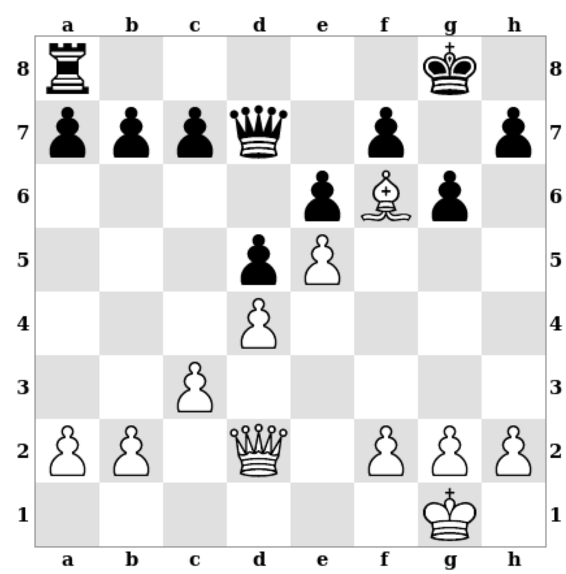
Weak Squares

Pawn structures can also create weak squares. These squares may make good homes for the opponent's pieces, which can lead to trouble.



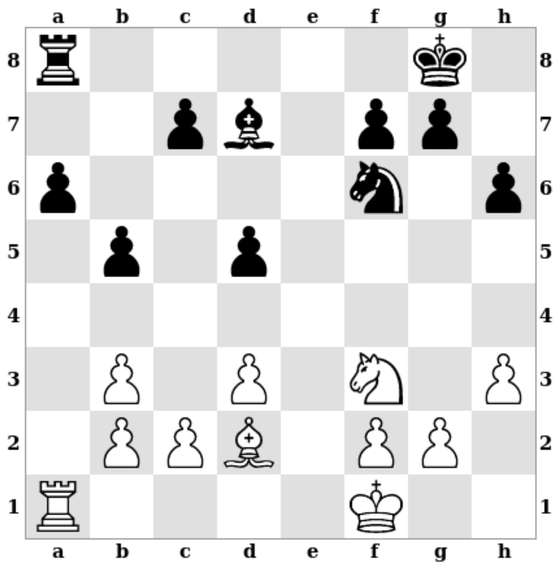
In this position, Black's e-pawn has advanced to e5 and his c-pawn has been traded off. As a result, he can't attack the d5 square with pawns, and so it has become weak. d5 is a good square for White's knight, which Black may find hard to get rid of.

In this position, Black's kingside pawns are all on light squares, and this has weakened the dark squares around his king (f6, g7, and h6). White's bishop already occupies f6, and White's queen is ready to make use of h6. In fact, White can mate in two moves with 1.Qh6 and 2.Qg7#.



Piece Activity

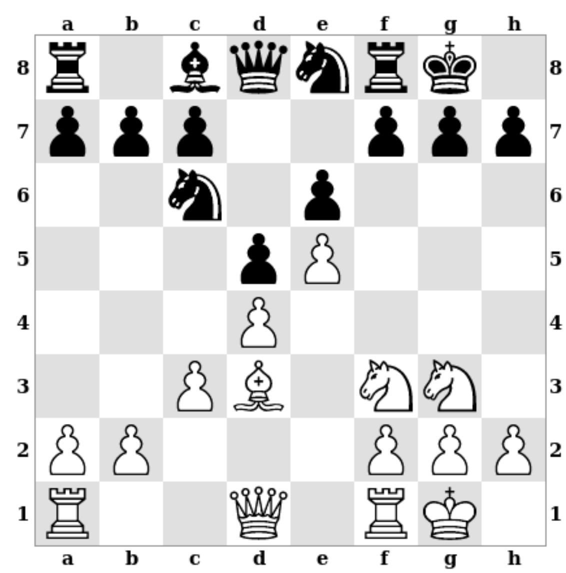
Pawns often get in the way of other pieces, so pawn structures can affect piece activity. In the starting position of a chess game, for example, none of the pieces other than the knights can move without having pawns move out of the way first.



Here White has doubled pawns on b2 and b3, which has given his rook an *open file* (the a-file). As a result, White's rook is more active than Black's (it has more legal moves), and it attacks one of Black's pawns, while Black's rook is only defending. Since it is also difficult for Black to attack White's doubled pawns, they are really a strength of White's position, rather than a weakness.

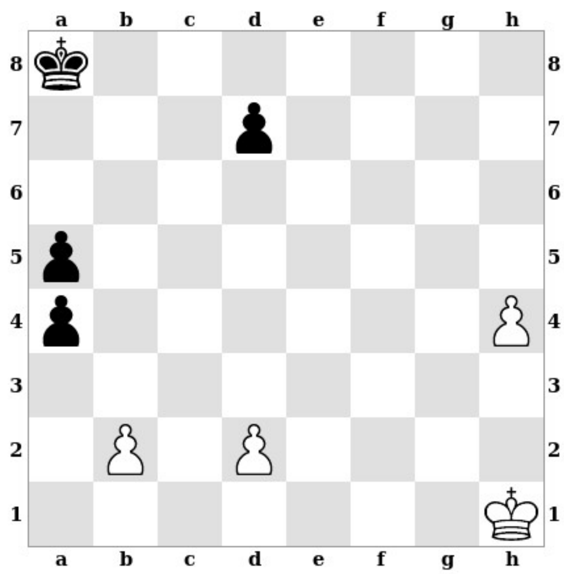
This example illustrates two very important points. First, pawns are not necessarily a serious weakness just because they fall into one of the previously mentioned categories of weak pawns. Second, weak pawns often bring some corresponding advantages. It is especially common for doubled, isolated, and hanging pawns to come with open files that rooks and queens can make use of.

In this position, White's bishop is much more active than Black's. The difference is in the players' pawn centers: Black's center pawns are on the same color squares as his bishop, blocking it in and making it a *bad bishop*, while White's center pawns are on the opposite color to his bishop, allowing it to move freely.



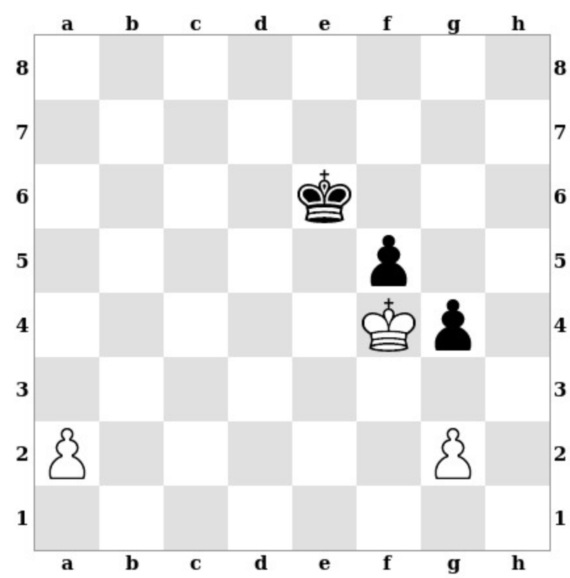
Passed Pawns and Pawn Majorities

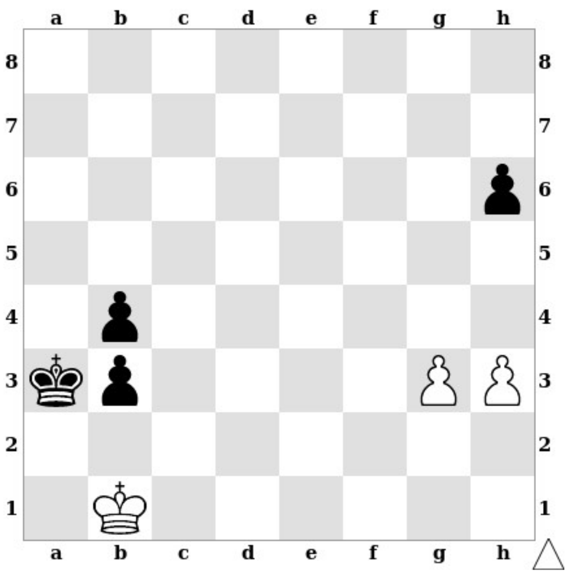
Pawn structures can also affect pawn promotion. In some cases the pawn structure may allow a player to promote a pawn by force, while in others, a player's pieces may become less active while trying to prevent a pawn from promoting.



A *passed pawn* is a pawn that doesn't have any enemy pawns stopping it from advancing to the eighth rank and promoting. In this position, White's h4 pawn is a passed pawn. White can win by simply pushing it up the board and promoting it. None of the other pawns in this position are passed pawns because they all have enemy pawns standing in their way.

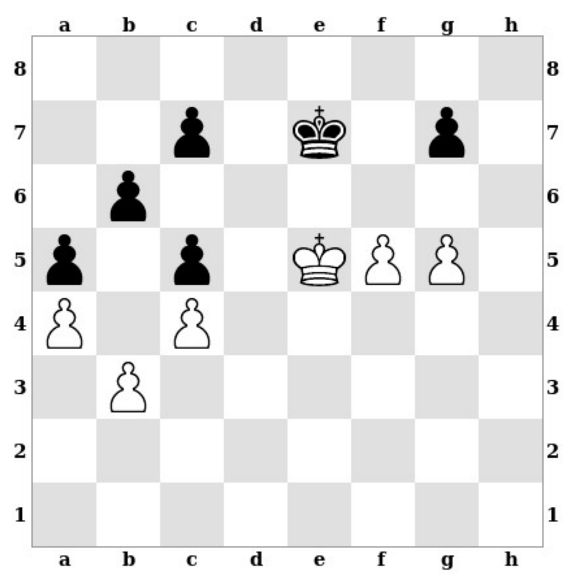
Here Black's king is close enough to White's passed a-pawn to catch it before it promotes. However, White still has a winning advantage since his a-pawn is an *outside passed pawn*: a passed pawn that is on the other side of the board, far away from the other pawns in the position. Black's king can stop the a-pawn from promoting, but he has to move away from his own pawns to do so, for example: 1.a4 Kd6 2.Kxf5 Kc5 3.Kxg4 etc.





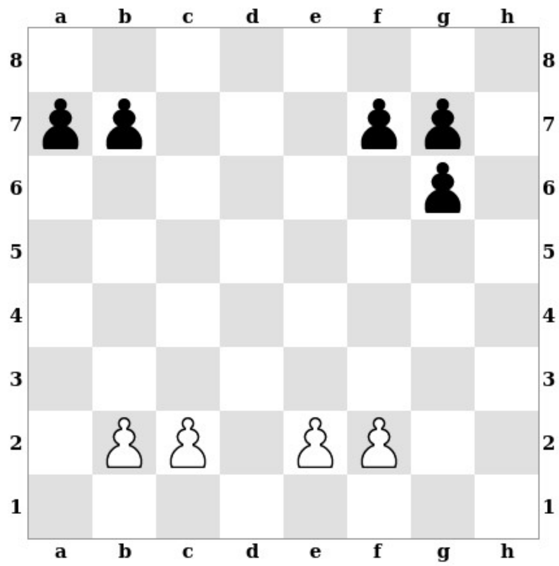
Here Black has passed pawns on b3 and b4, but White's king is preventing these pawns from promoting. White doesn't yet have a passed pawn himself, but he does have a two-to-one *pawn majority* on the kingside.

It is best to advance a pawn majority by moving the unopposed pawn first. White can win with 1.g4 followed by 2.h4, 3.g5, and then just pushing his g-pawn to promotion. It would be a huge mistake to begin with 1.h4? instead, since after 1...h5, White can't safely advance his majority any further.



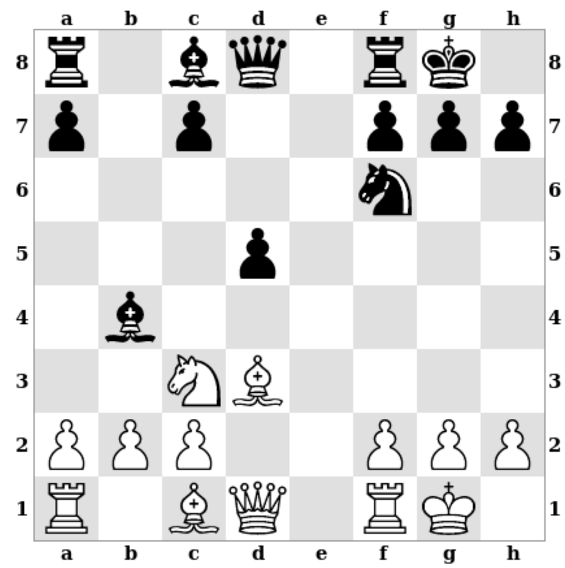
In this case, the two players have *rival* pawn majorities; White has a *kingside majority*, while Black has a *queenside majority*. White has a winning advantage, however, due to the fact that his majority is a *healthy majority*, while Black's majority is *crippled* by his doubled pawns. White wins by creating an outside passed pawn, using it to distract Black's king, and then running over with his own king to capture Black's pawns. For example: 1.f6+ gxf6+ 2.gxf6+ Kf7 3.Kf5 Kf8 4.Ke6 Ke8 5.f7+ Kf8 6.Kd7 etc.

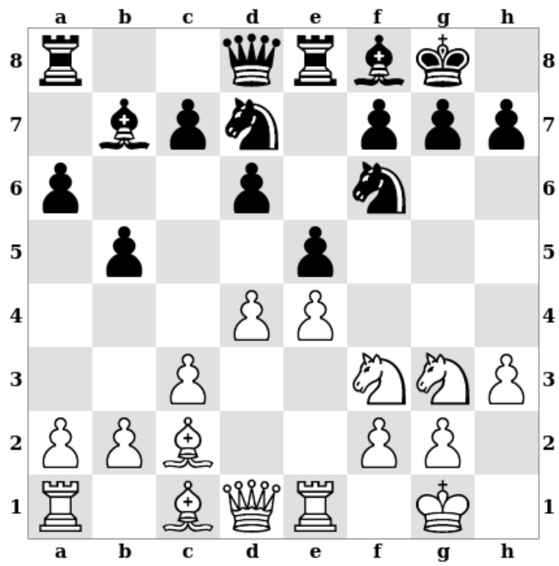
Pawn Centers



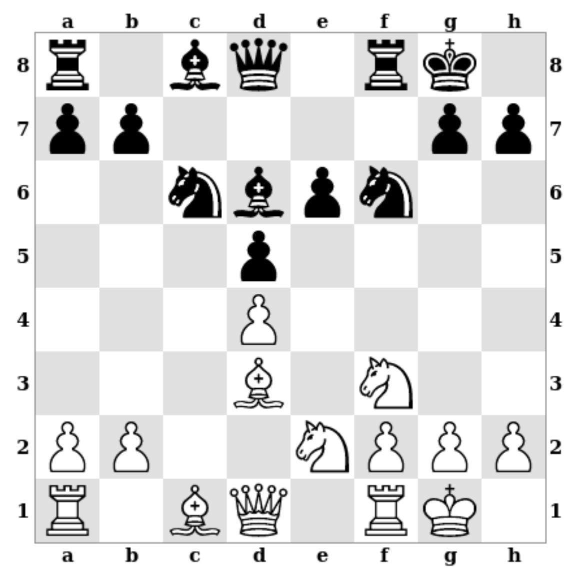
We can classify files on the board according to what combination of pawns are on them. In this position, the d-file and h-file are both *open files* since they have no pawns on them. The a-file, c-file, e-file, and g-file are *half-open files* since they have one player's pawns on them but not the other's. Finally, the b-file and f-file are *closed files* since they have pawns of both colors on them.

We often refer to entire positions as *open positions* or *closed positions*. Rather than trying to find an absolute rule to distinguish the two, it is more important to think about the degree to which any position is open or closed. This is determined by looking at all of the files on the board but especially the two central files (the d-file and e-file). The position shown here is quite open, for example, because there are both one fully open and one half-open file in the center.





By contrast, this position is closed since there are no open or half-open files anywhere.



And, of course, there are some positions that don't fall clearly into one category or the other. We might call this a half-open position, though the exact label we use is not all that important.

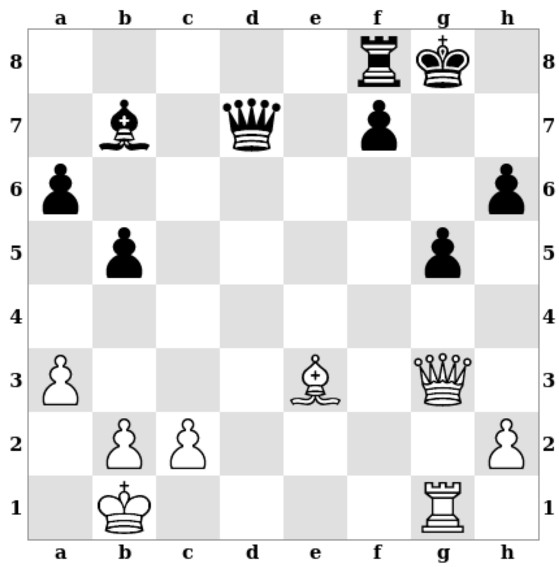
As we will see in further examples, there are many strategic ideas and guidelines that apply more to open positions than to closed positions and vice versa.

4 – King Safety

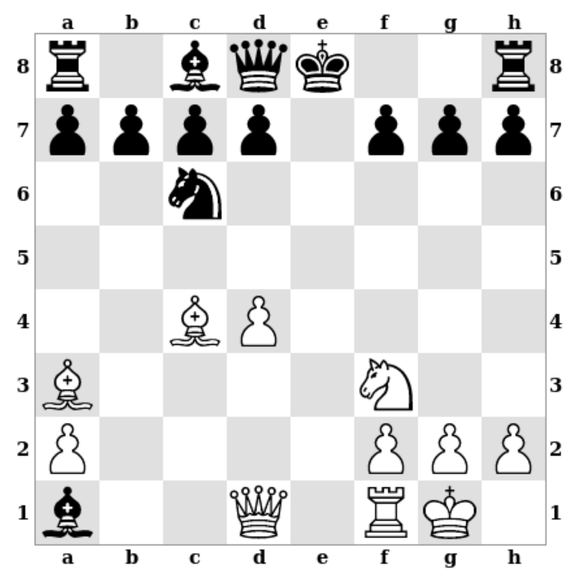
The king is the most valuable piece in the game, but it moves slowly and has trouble defending itself. For these reasons, the safety of the king can become a long-term issue in many positions. King safety is mostly a matter of how easily the king can be attacked and how easily it can be defended. However, there are a few factors that can serve as clues as to whether the king is safe or not.

Pawn Cover

In the opening and middlegame, the king is usually safest when it has some pawns covering it to make it harder for enemy pieces to attack it. When the king has no pawn cover, it can quickly find itself in danger.

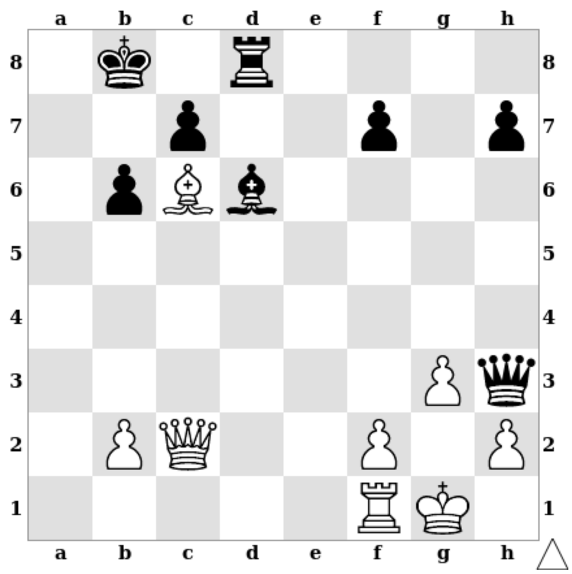


In this position, Black has advanced the pawns in front of his king recklessly. White can exploit this with the move 1.Bxg5, destroying the king's pawn cover completely. If Black captures the bishop, he gets mated right away: 1...hxg5 2.Qxg5+ and 3.Qg7#. Black should decline the sacrifice, though he is still losing, e.g. after 1...f5 2.Bxh6+, when White has a material advantage and a continuing attack.

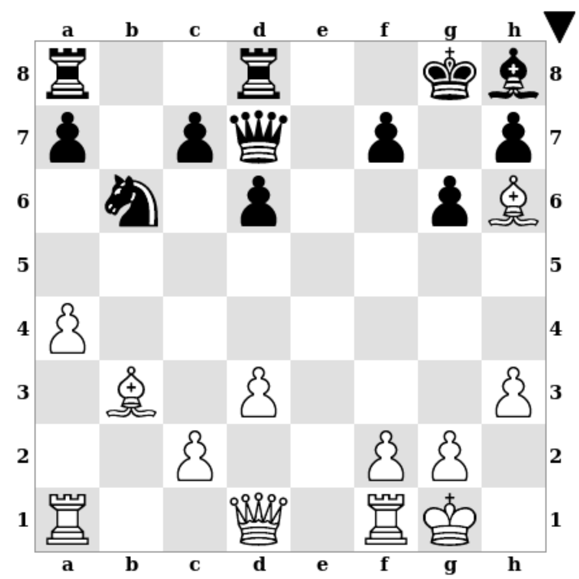


Lack of pawn cover for the king can quickly become an issue in the opening phase, especially in open positions. For this reason, it is often a good idea to castle early in the game if it looks like the position could become open. Here, Black has delayed castling too long and the e-file has become open. White can punish him with 1.Re1+, when Black has nothing better than 1...Ne7 2.Bxe7, losing his queen.

Weak Squares



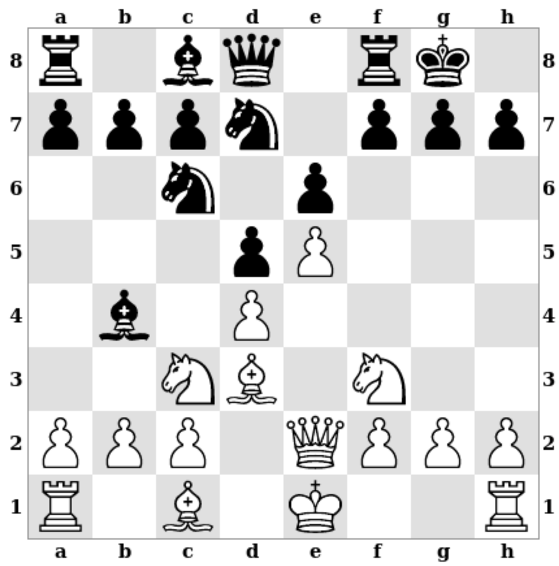
Kings can also find themselves in danger when there are groups of weak squares around them. In this position, both castled king positions have a system of weak light squares around the king. However, because only White still has a light-square bishop, he is better prepared to exploit Black's weak squares than Black is to exploit his. White can force checkmate with 1.Qa4, threatening Qa8#. Black can only delay mate by sacrificing his queen.



Weak squares around a king are often less of an issue when there is a piece available to defend them. It is especially common for a bishop to help protect weak squares of a particular color. Here Black's kingside has a system of potentially weak dark squares, but he isn't currently in much danger since he still has his dark-square bishop. However, he could easily lose by giving up this bishop, for example after 1...Bxa1? 2.Qxa1, when there is no way to stop 3.Qg7#.

Piece Activity

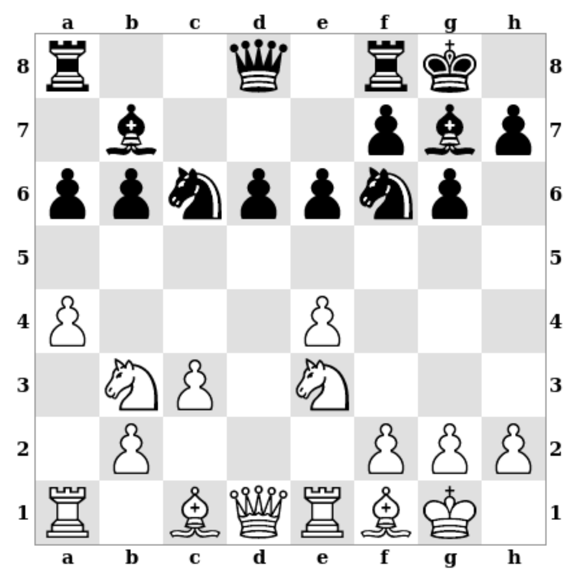
At a more sophisticated level, king safety is affected by the activity of both sides' pieces. When the attacker has more active pieces near the king than the defender, the king may be in danger in spite of having good pawn cover and no obviously weak squares nearby.

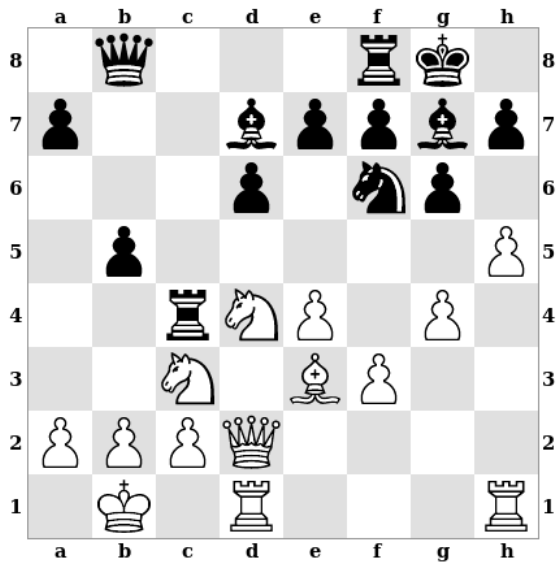


In this position, Black has just castled. His king position might look quite safe since he has three unmoved pawns in front of his king. However, White has a space advantage on the kingside and a number of pieces with easy access to that part of the board. By contrast, Black has no pieces that are on the kingside or able to get there quickly. White's e5 pawn also interferes with Black's pieces; for example, it stops a black knight from coming to f6. As a result, White can start a winning attack with the famous "Greek Gift" bishop sacrifice: 1.Bxh7+ Kxh7 2.Ng5+. The further details are somewhat complicated, but White's attack should lead to checkmate or winning large amounts of material quite soon.

Opposite-Side Castling

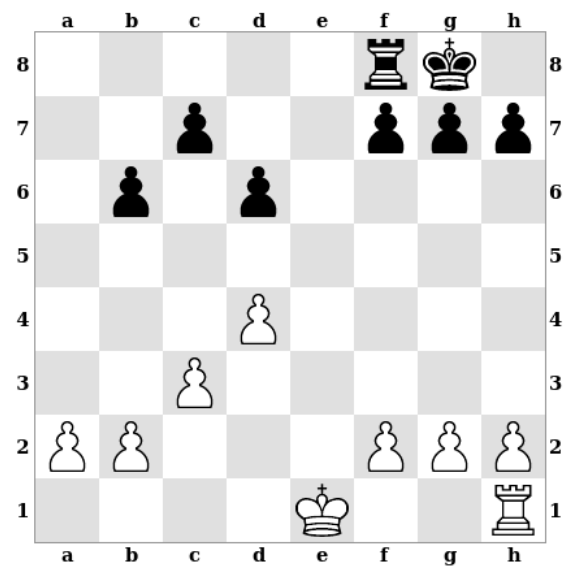
King safety is also sometimes affected by whether or not the players choose to castle on the same or opposite sides from one another. This position comes from a line of the Sicilian Dragon in which both players have castled kingside. Because the kings are castled on the same side of the board, neither player can begin a pawn storm against his opponent's king without moving the pawns in front of his own king. As a result, the players have adopted more cautious strategies that don't revolve around immediate kingside attacks.





This position also comes from the Sicilian Dragon, but here White has chosen a more aggressive strategy in which he castles queenside. With the kings castled on opposite sides of the board, both players are free to use pawn storms against each other's kings without exposing their own kings in the process. For this reason, opposite-side castling often leads to a race between two kingside attacks, which puts both kings in increased danger.

The Endgame



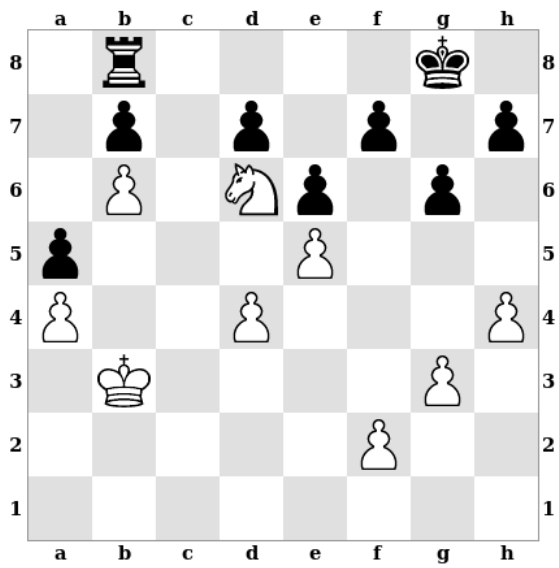
In the endgame, it is often quite safe for the king to come out in the open, since there are not enough pieces left on the board to put it in real danger. Once it is safe to do so, bringing the king into the game is quite important since the king can play a valuable role as an aggressive piece itself.

Here White should play 1.Kd2, bringing out the king and preparing to activate his rook. Instead, 1.0-0? would be a serious mistake, allowing Black to play 1...Re8 and 2...Re2.

5 – Piece Activity

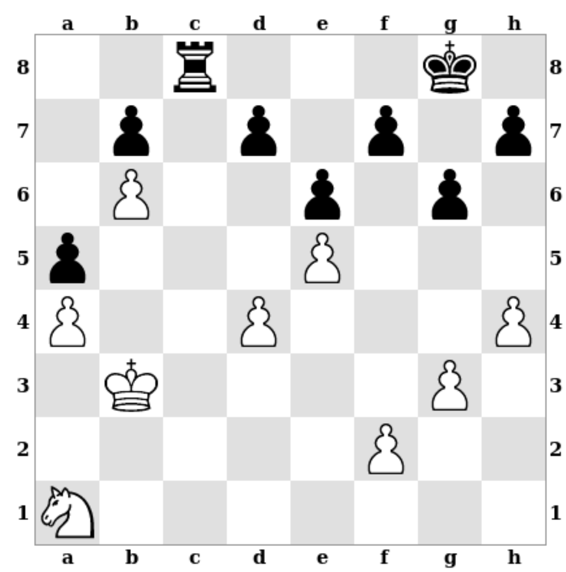
Because some pieces are stronger than others we give them material values (e.g. a rook is worth about five pawns, while a knight is worth about three); but the real value of a piece in a specific position may be quite different from what its material value would suggest. Some pieces are *active*, well-placed and doing useful things, while others are *passive*, poorly placed and unable to do anything useful. The factors that make pieces active and passive are closely tied to the way that different chess pieces move. For example, a knight becomes active under different conditions than a rook does.

Active and Passive Pieces



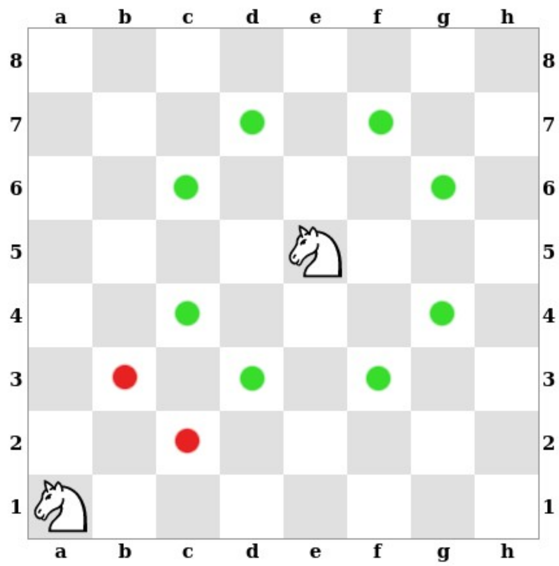
In this position, White is down material (he only has a knight relative to Black's rook). However, his knight is extremely active and impossible for Black to dislodge from its position, while Black's rook is in a passive position and cannot easily be activated. In fact, White has an easily winning advantage. He can continue with Kc4-b5-xa5 which Black has no good way of defending against. Black would love to play ...Rc8 at some point, but White's well-placed knight would simply capture the rook.

This position features the same material balance and the same pawn structure as the one above. However, Black's rook is now very active on the open c-file while White's knight is passively placed on a1. This change results in a completely different evaluation of the position. Black is now winning, one idea for him being ...Rc6-xb6.



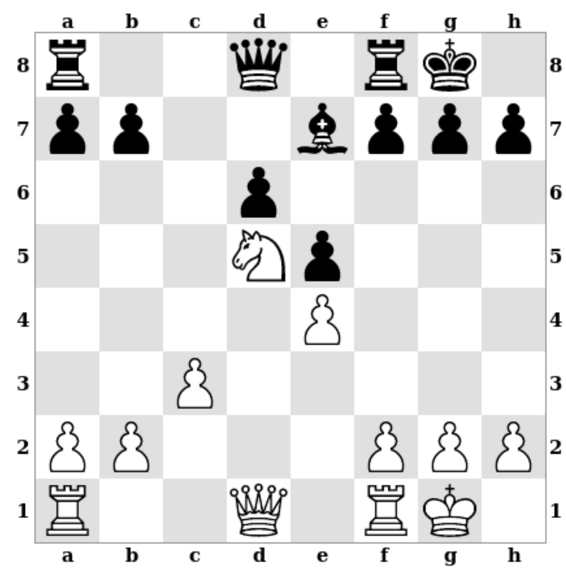
Knights

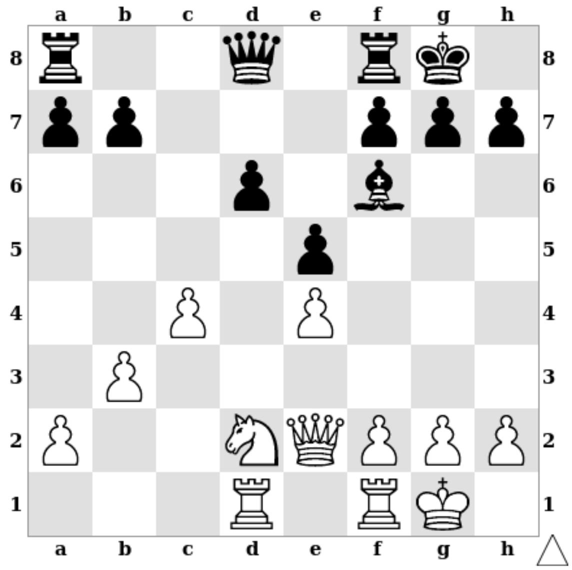
Knights are short range pieces; they do well when they are close to important parts of the board, such as the center or the opponent's territory.



A knight in the center of the board attacks eight squares, while a knight on the edge normally attacks four squares, and a knight in the corner only attacks two. Clearly, knights are more active closer to the center of the board.

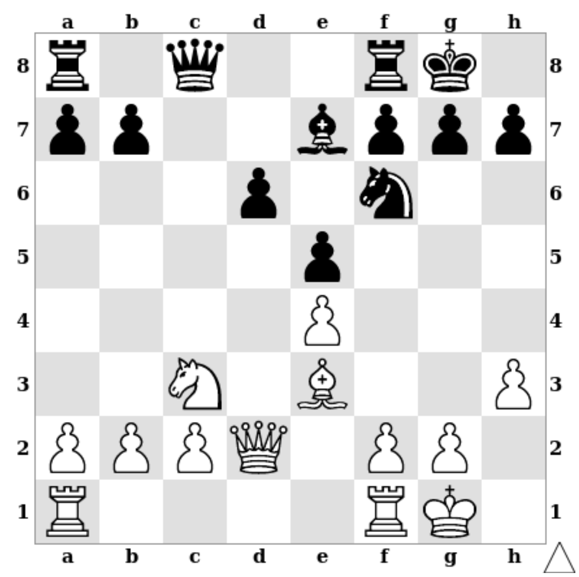
One of the most active positions for a knight is an *outpost*. An outpost is a square near the opponent's territory that the opponent cannot attack with pawns. In this position, d5 is an outpost since Black's c-pawn has been traded off and his e-pawn has advanced too far to attack d5. This particular outpost is made even stronger by the fact that Black doesn't have any knights or a light-squared bishop that he could try to trade for White's knight.

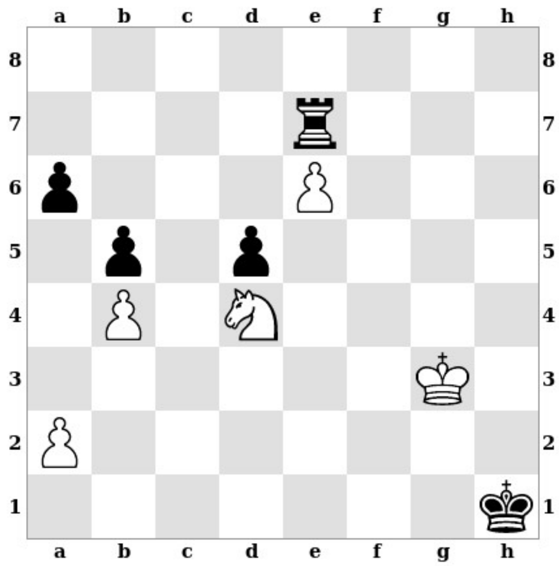




It is sometimes possible to re-route a knight to an outpost square. For example, White's best move here is probably 1.Nb1. At first this move looks counterproductive since it moves the knight away from the center. However, the idea is to play 2.Nc3 and 3.Nd5, putting the knight on a much better square than d2. Black has no good way to stop this and White ends up with a strong knight outpost.

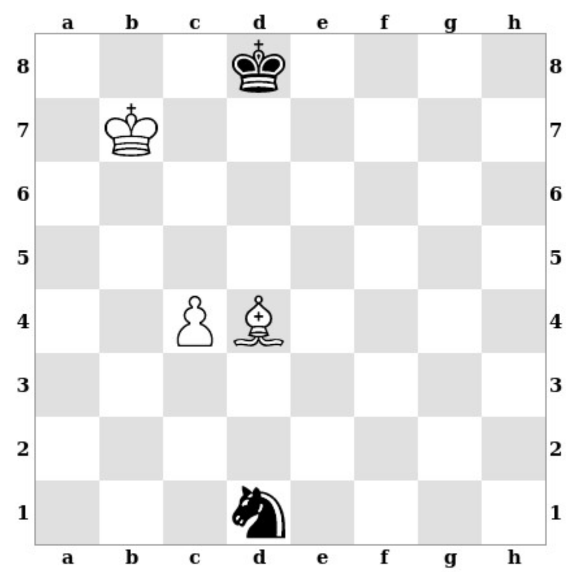
It is also sometimes necessary to play some preparatory moves to secure a knight outpost. White would love to establish his knight on d5 here, but the immediate 1.Nd5 is met by 1...Nxd5. Instead, White should play 1.Bg5 and 2.Bxf6 in order to eliminate Black's knight and only then play 3.Nd5.





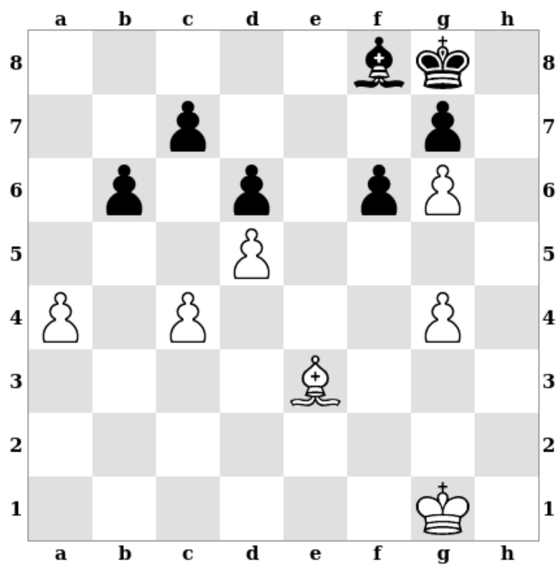
Because knights can jump over other pieces, they are also quite good at *blockading* pawns (standing in their way to stop them from advancing). In this position, White's knight and Black's rook are each blockading a pawn. However, White's knight is still quite active, attacking the black pawn on b5 and guarding White's e6 pawn. Black's rook, by contrast, has been left rather passive by its role as a blockader. White has a strong plan in Kf4-e5-xd5-d6 followed by trying to promote his e6 pawn.

Knights often get into trouble on the edge of the board. Here Black's knight is stuck on the edge since White's bishop controls all the squares it can move to. Without the help of the knight, Black has no way to stop White's pawn. White can win by simply moving his pawn up the board and promoting.



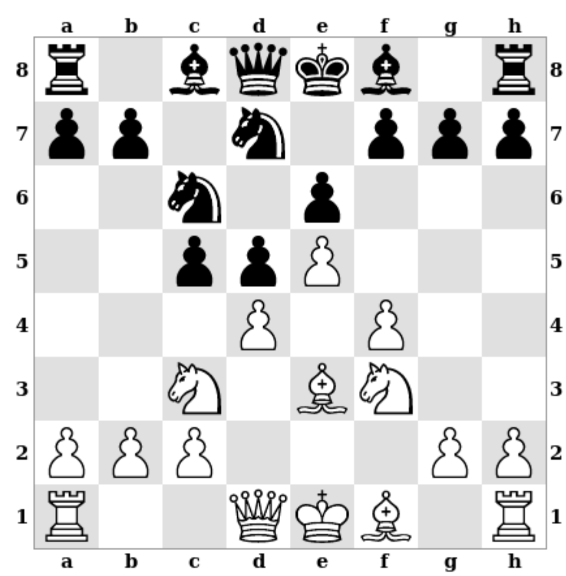
Bishops

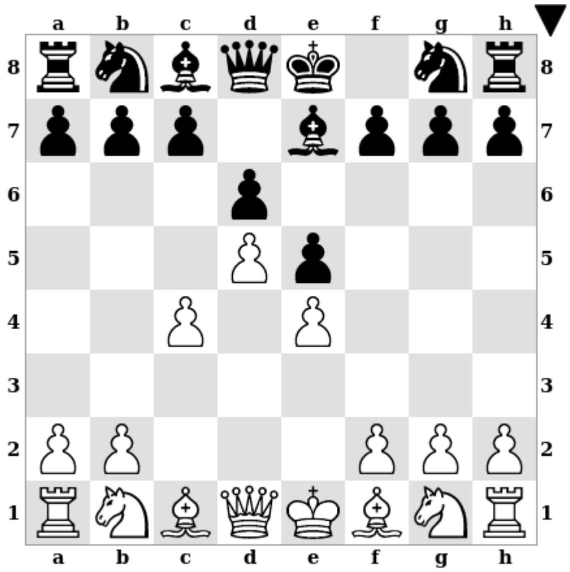
Bishops are long-range pieces. They don't necessarily need to be in the center to be active, but they need open diagonals to move along.



Bishops are most active when they are not blocked in by their own pawns. Here Black's bishop is quite passive since his pawns are on the same color squares as those the bishop moves on. White's bishop is active since his pawns are on the opposite color to his bishop.

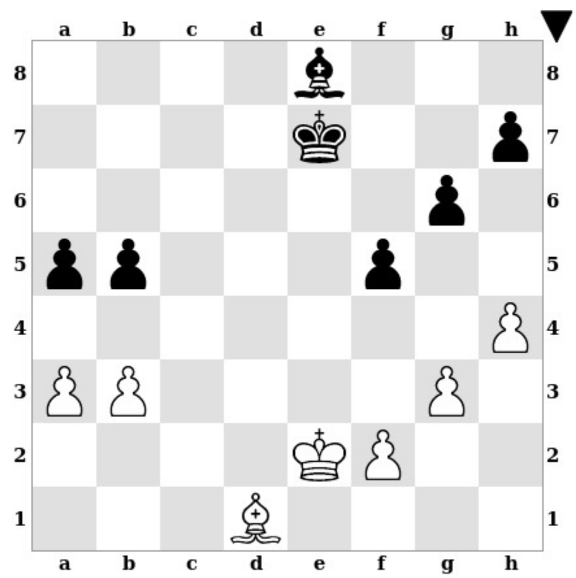
Many opening variations lead directly to pawn structures in which bishops become passive. This position comes from a popular variation of the French Defense. Black's center pawns are fixed on light squares, so his light-square bishop will be passive for the foreseeable future. If Black could remove both players' light-square bishops from the board, his position would immediately become much better.





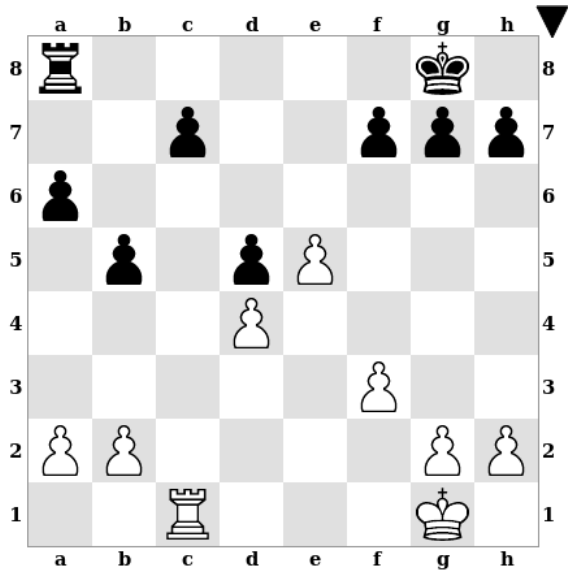
The distinction between good and bad bishops can often be used to decide which pieces to trade and which to keep on the board. In this position, the central pawn structure is already well defined and has clear implications for the bishops. White's pawns are on light squares, so his dark-square bishop is his better bishop, while Black's pawns are on dark squares, so his dark-square bishop is his worse bishop. These observations suggest a strong idea for Black: with the move 1...Bg5 he can try to force a trade of his bad bishop for White's good bishop. White can only avoid this by playing 2.Nd2, but after 2...Nd7 3.Nf3 Bh6, Black's bishop will wait on h6; when White's knight eventually moves, Black will carry out the bishop trade anyway.

A player can sometimes make his own bishop better than his opponent's by choosing the right pawn structure. If Black moves first here, he should play 1...b4. This both fixes his own queenside pawns on the opposite color to his bishop and fixes White's pawns on the same color as his bishop. Black's bishop will be more active than White's as a result. This will also make White's queenside pawns more vulnerable than Black's. If Black instead misses his chance with 1...Kd6?, then White should respond with 2.b4 himself, seeking the same advantages.



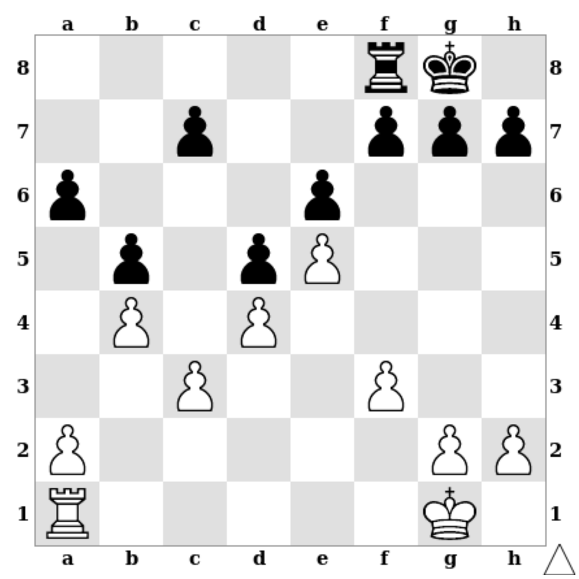
Rooks

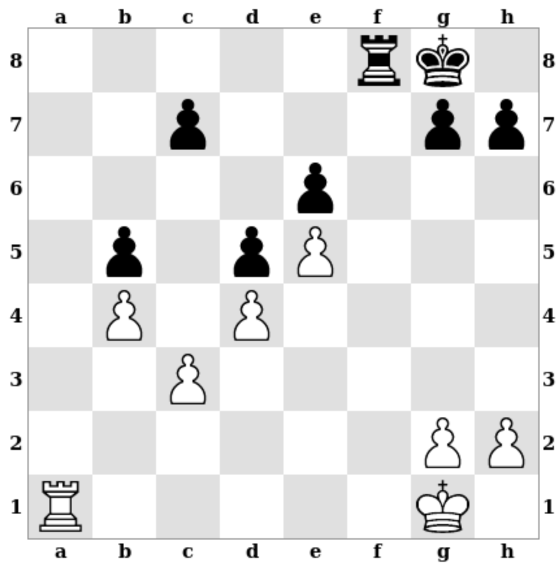
Like bishops, rooks don't need to be centralized to be active, but they do require open lines to move along.



Rooks are most active on open files. Here White's rook is on the open c-file, while Black's rook is on the closed a-file. As a result, White's rook is much more active than Black's.

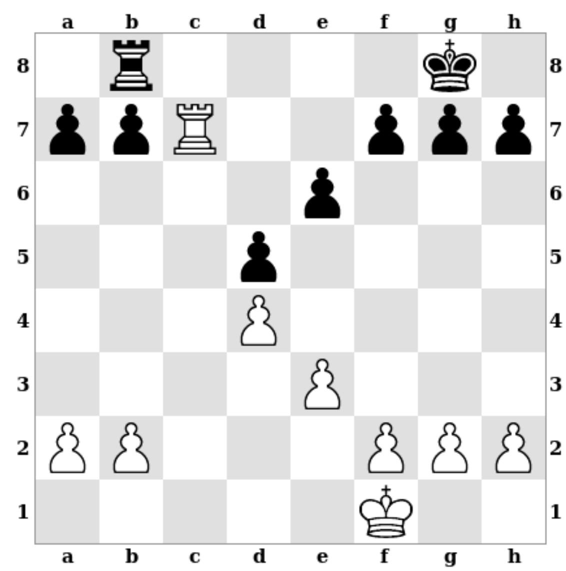
It is sometimes necessary to create an open file for a rook. The most common method of doing this is to use a *pawn break*, a pawn move that attacks an opponent's advanced pawn. Here White can open the a-file for his rook by playing 1.a4, intending 2.axb5. If Black responds with 1...bxa4, then White plays 2.Rxa4. In either case, White's rook will now stand on an open file which it can use to invade Black's position or attack Black's pawns.

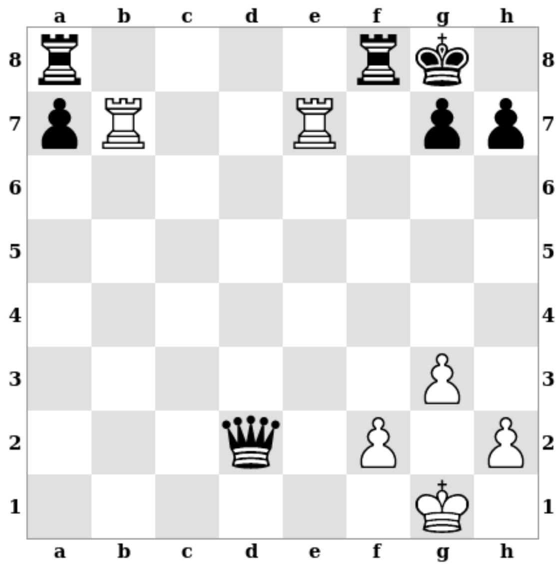




The actual value of an open file depends on whether or not it can be used to attack the opponent's pawns or infiltrate his position. In this example both rooks are on open files, but White's open file is much more valuable. The f-file isn't really very useful to Black's rook because White's king and pawns cover all of the rook's entry points into White's position. By contrast, White's rook can easily use the a-file to get into Black's position and start causing trouble.

Rooks also become very active on the seventh rank (or the second rank for black rooks). Here White's rook has reached the seventh rank and now dominates the position. Black's king can't approach the rook easily since this would let the rook capture Black's kingside pawns. Black's rook also can't move without hanging the b-pawn, and moving the b-pawn itself would hang the a-pawn. Black is left without good moves and White can gradually improve his position.



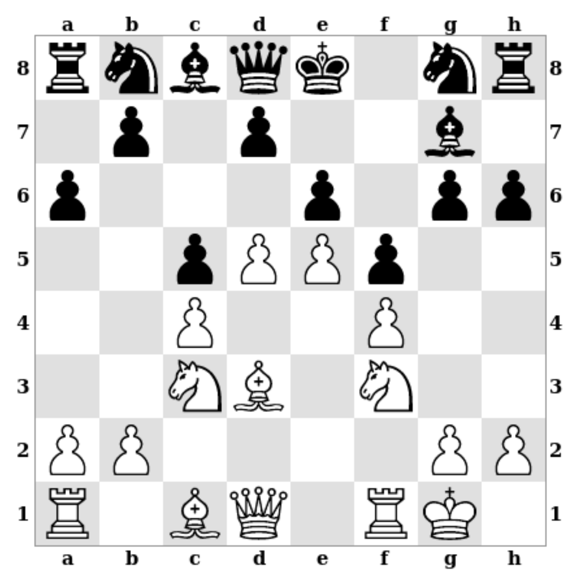


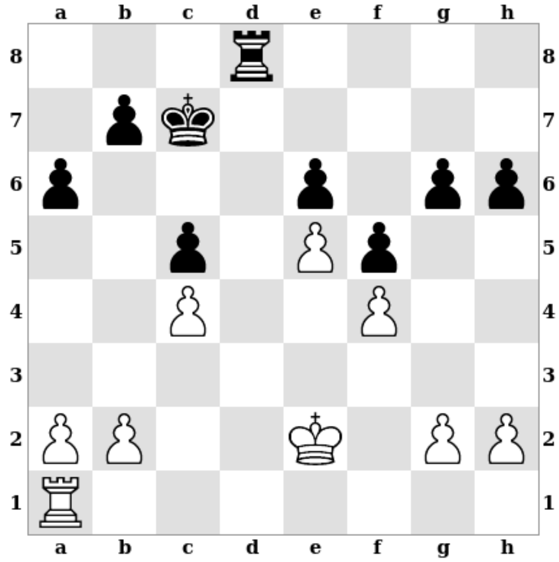
Two rooks working together on the seventh rank can be especially powerful. Here White's seventh rank rooks can even force a quick mate: 1.Rxg7+ Kh8 2.Rxh7+ Kg8 3.Rbg7#

Space

In some positions, one of the players has a *space advantage*, meaning that he controls more territory than his opponent. There is no simple way to measure space, but it is closely related to how far advanced a player's pawns are. Having more space typically gives you more room for your pieces and makes them more active.

Here White's advanced pawn center gives him a space advantage, which has made his pieces more active than Black's. For example, it is difficult to imagine good placements for both of Black's knights or to see how he will be able to develop his light-square bishop effectively. White's pieces, by contrast, have many good squares available to them.

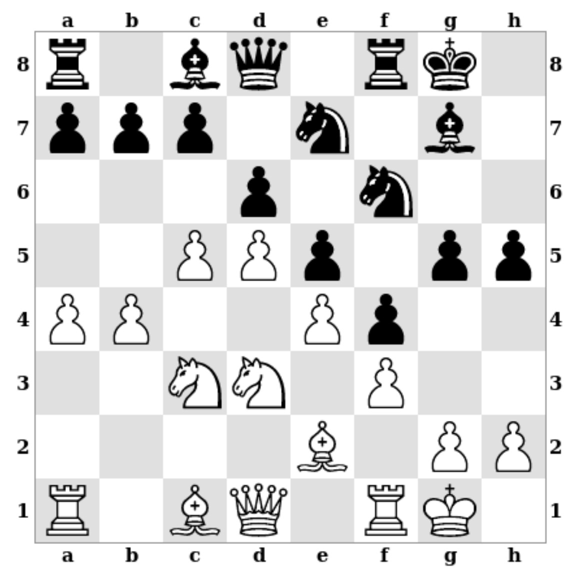


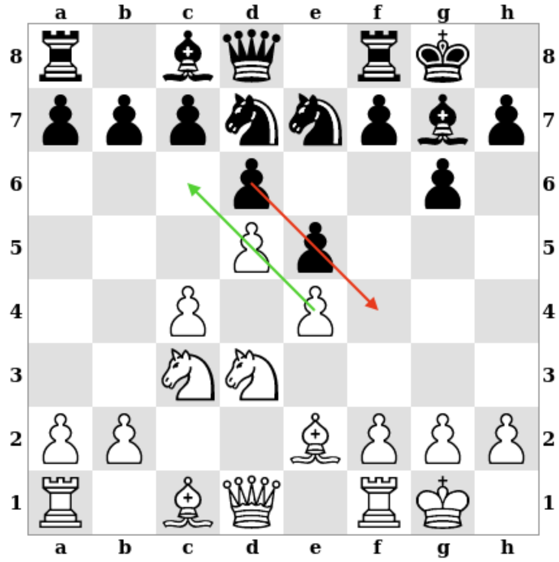


A space advantage is most important when there are many pieces on the board and becomes less important the fewer pieces there are. In this position White has a space advantage, but with so few pieces left it just doesn’t matter much.

This suggests that the player with a space advantage should avoid trading pieces, while the player with less space should trade off some pieces to make his position easier.

In some positions, the two players end up with rival space advantages in different parts of the board. In this position, arising from a line of the King’s Indian Defense, White has a space advantage on the queenside, while Black has a space advantage on the kingside. Each player will find it easier to attack on the side of the board where he has more space.



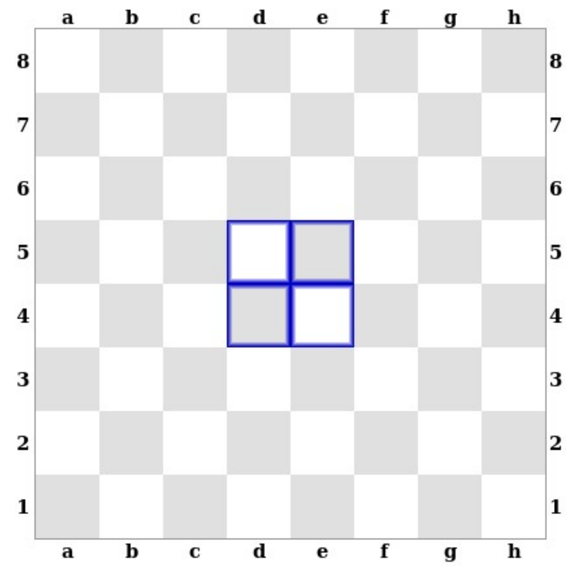


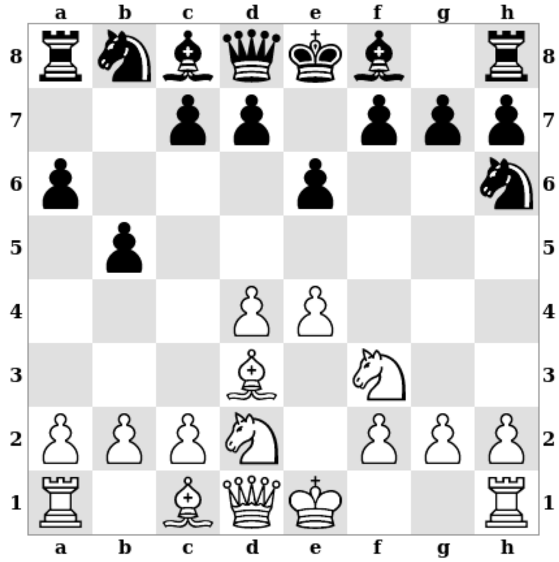
This position occurs at an earlier point in the same variation of the King's Indian. Even though the players have not begun their pawn storms yet, it is already clear which side of the board each player should focus on. White's more advanced center pawn is closer to the queenside, so he should play there, while Black should play on the kingside where his more advanced center pawn is.

A useful idea here is the *pawn pointing rule*: when the center pawns become locked, each player should focus his play on the side of the board that his center pawns "point" toward. In this position, the white pawns on e4 and d5 point toward the queenside, while the black pawns on d6 and e5 point toward the kingside.

Central Control

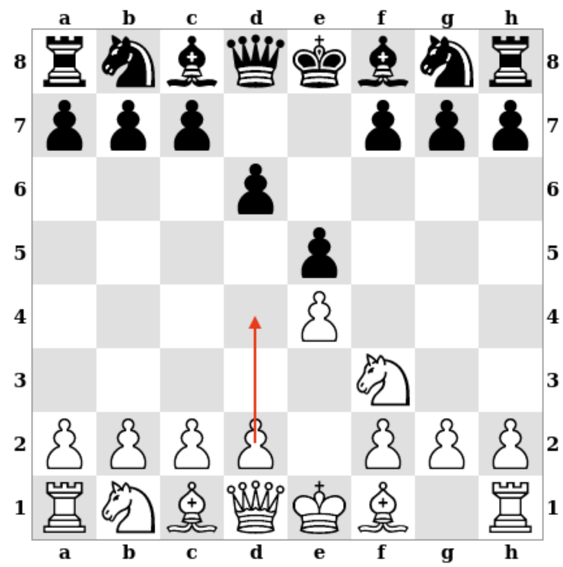
The center of the board is generally the most important part of the board for at least two reasons. First, the center is close to every other part of the board. If you want to move pieces from one part of the board to another, it helps to be able to move them through the center. Second, some pieces, like knights, are most active when they can occupy squares in the center of the board. For these reasons, the player with better control of the center will often have an advantage as a result.





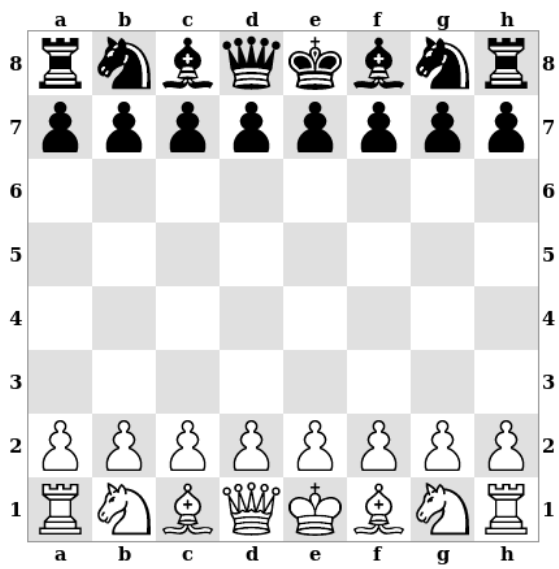
In this position, White has greater central control than Black. White has multiple pieces attacking both e4 and e5, and additionally has pieces attacking d4 and d5. By contrast, Black only has one piece attacking d5 and no other pieces attacking the center at all. As a result, White's pieces already have more good squares available to them than Black's do.

Increasing your own control of the center is sometimes accomplished by trying to reduce your opponent's central control. This is another situation in which pawn breaks are useful. In this position, which comes from Philidor's Defense, White should start attacking Black's pawn center by playing 1.d4. This threatens to win a pawn on e5 and encourages Black to give up his strong central pawn, which would leave White with better central control.



6 – The Opening

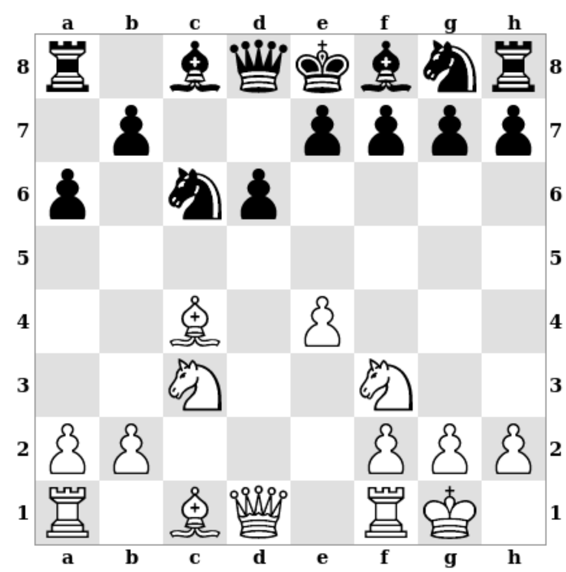
The first phase of a chess game is known as the *opening*. The strategic ideas in the opening are largely the same as in the middlegame, but the emphasis is different because of the unique nature of the starting position. The best way to begin thinking about opening strategy is to think about what makes the starting position so unique.

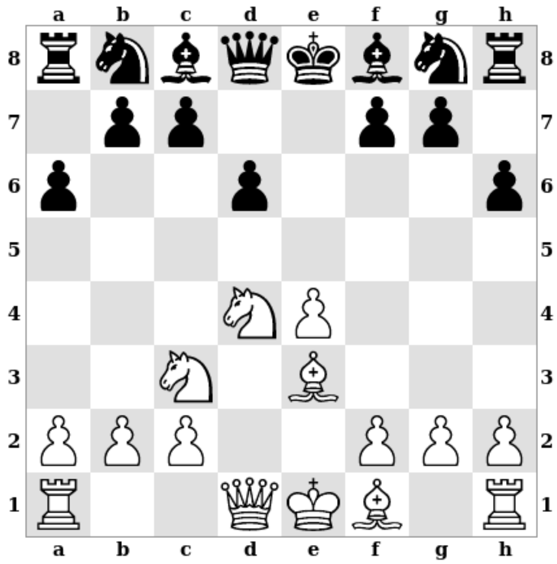


Compared to a normal middlegame position, there are two important things that stand out here. First, both players' pieces are totally inactive. Second, neither player has any space or control of the center. The most important strategic goals in the opening come from addressing these points. Both players will try to *develop* their pieces quickly to make them more active. At the same time, the players will focus on controlling the center of the board and gaining space where possible.

Development

The most important goal in the opening is *development*, the process of bringing pieces out to more active squares. In this position, from a line of the Morra Gambit, White has sacrificed a pawn in order to develop his pieces more quickly. White now has three pieces developed, has castled, and will not need another pawn move to develop his second bishop. Black, by contrast, has only one piece developed, has not castled, and will need another pawn move in order to develop his dark-square bishop. White's lead in development gives him good *compensation* for his sacrificed pawn.



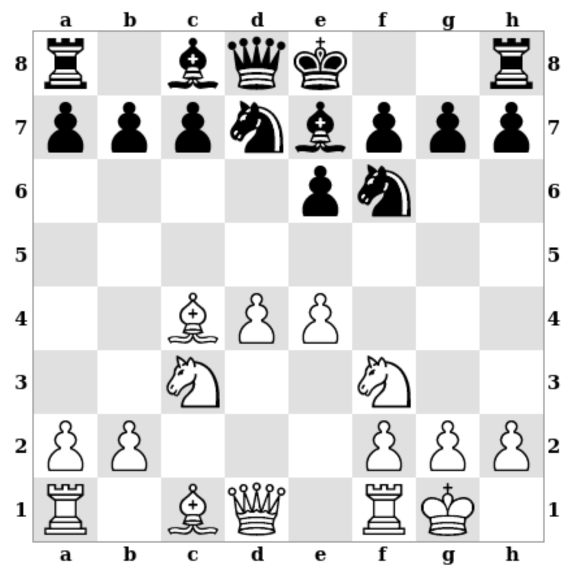


Pawns can’t achieve the level of activity of other pieces because their movements are so limited. For this reason, pawn moves can’t really be considered developing moves on their own. In this position, for example, White has developed three pieces and Black has developed none, giving White a significant lead in development.

Of course, pawn moves do sometimes contribute to development by opening lines for bishops, rooks, or the queen. In fact, without making any pawns moves in the opening, it would be impossible to develop any pieces other than the knights.

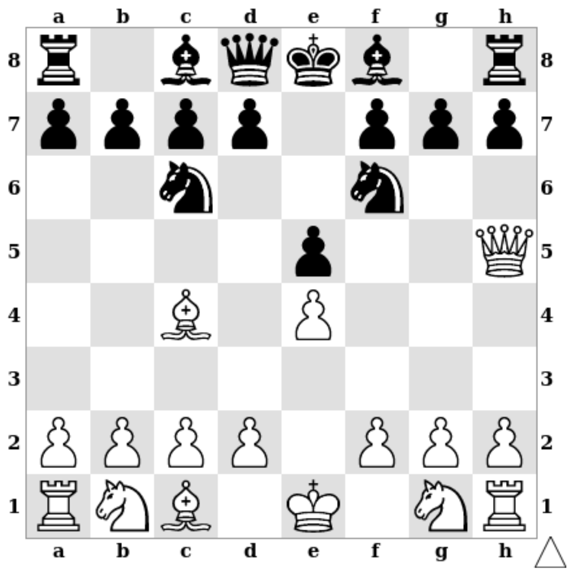
Central Control

The battle for control of the center begins directly in the opening. This position comes from a line of the Queen’s Gambit Accepted, in which Black has captured White’s c-pawn, giving up one of his center pawns in the process. Over the next moves, White has established both his center pawns on the fourth rank and developed a number of pieces to active positions from which they attack the center. Black has played too passively and has not contested White’s central control, giving White a pleasant advantage.



King Safety

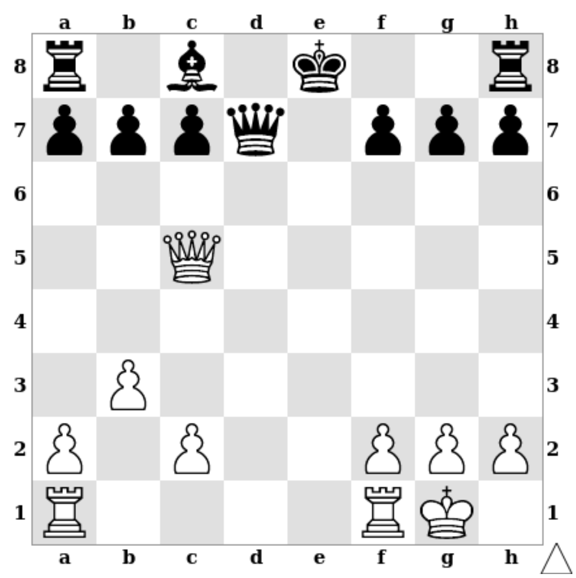
In the initial position, the kings both have plenty of pawn cover and appear to be quite safe. However, this is misleading for two reasons.

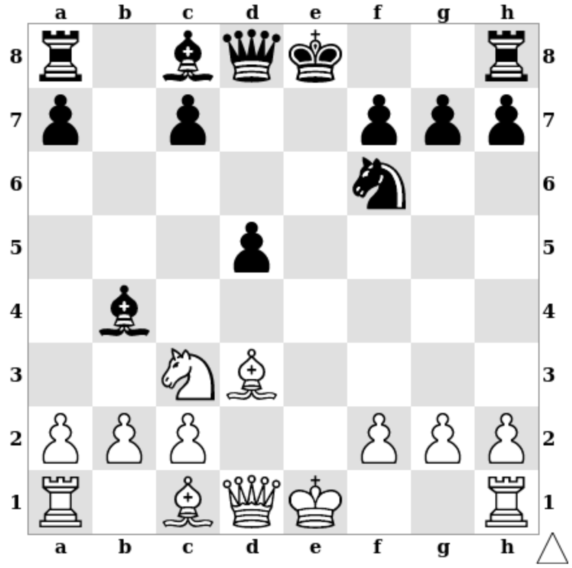


First, the squares f7 and f2 (for black and white, respectively) are very vulnerable in the opening. These squares are right next to the two kings and are not guarded by any pieces other than the kings. This makes them easy targets for a quick attack.

This position comes from a version of what is known as *Scholar's Mate* or the *Four Move Checkmate*. The game began 1.e4 e5 2.Qh5 Nc6 3.Bc4. Black then blundered with 3...Nf6??, which now allows White checkmate in one move with 4.Qxf7#. Black should instead have played something like 3...g6 4.Qf3 Nf6, when he would have successfully defended the attack. While this opening shouldn't work against a good defense, it highlights the weakness of the f7 square in the opening.

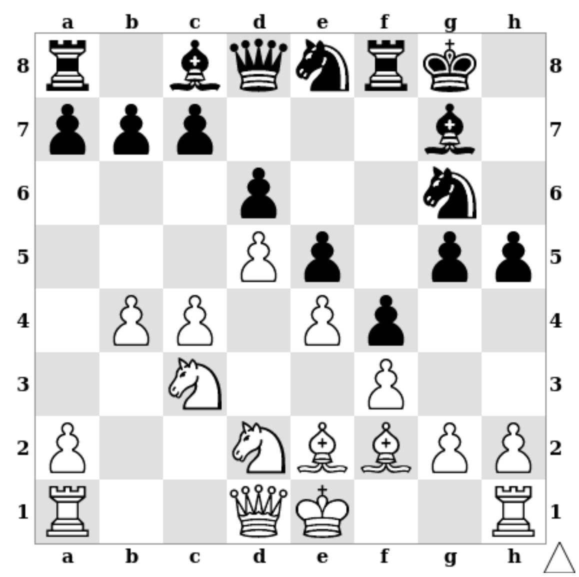
The second issue is that as the players fight for control of the center, the center pawns will often be traded, leaving open lines in the center. The kings may find their pawn cover suddenly missing, which can lead to trouble. Here Black has kept his king in the center too long. White can punish this by playing 1.Rfe1+ Kd8 2.Rad1, winning Black's queen for a rook.





One standard way to protect the king is to castle, since this moves the king away from open lines in the center. An added benefit is that castling brings one of the rooks closer to the center, which can help that rook become more active.

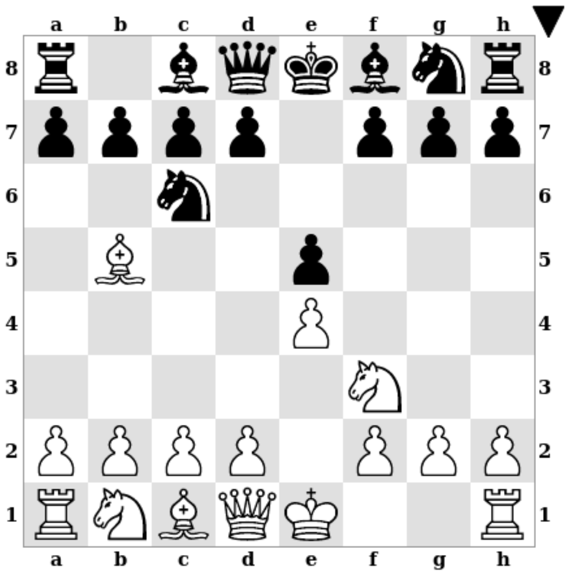
This position comes from a line of the Scotch Four Knights. The e-file has been opened and so a very common continuation here is for both players to castle: 1.0-0 0-0.



However, castling is not always a good idea and should not be done automatically. In this position it would actually be a terrible mistake for White to play 1.0-0. First of all, the center is closed and isn't likely to be opened any time soon, so White's king isn't really in danger on e1. Secondly, Black has already made a lot of progress on an aggressive kingside pawn storm, so White's king would be in serious danger after castling kingside.

Opening Theory

Chess players give names to specific sequences of moves beginning from the initial position. Certain sequences of moves are understood to be good for White, good for Black, or to give both players equally good positions. *Opening theory* refers to the chess world's collective knowledge of these sequences of moves and their assessments.



For instance, this position occurs after the moves 1.e4 e5 2.Nf3 Nc6 3.Bb5. This opening is known as the Ruy Lopez or the Spanish. It has been one of the most popular openings throughout chess history and has accumulated a large amount of theory.

Because White moves first in chess he begins the game with a small advantage. Opening theory has traditionally been based on the assumption that an opening needs to maintain or increase White's advantage in order to be a good opening for White. In order for an opening to be considered good for Black, it would need to lead to a position in which neither side has much of an advantage. In this latter case, Black is said to have *equalized*. As chess has been studied more and more deeply, it has become harder and harder to find meaningful advantages for White when Black plays well. As a result, modern opening preparation is as likely to involve White simply trying to get into a position that he knows well or enjoys playing, even if it is fairly equal, as to look for a lasting advantage out of the opening.

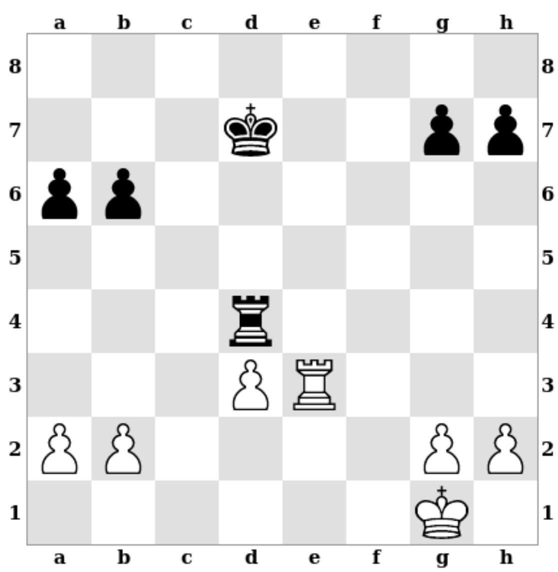
7 – The Endgame

The final phase of a chess game is known as the *endgame*. The endgame is characterized by the fact that there are fewer pieces on the board than in the opening or middlegame. An endgame typically does not involve more than a few pieces beyond the kings and pawns, though there is no absolute rule about how many pieces must be left before an endgame is reached.

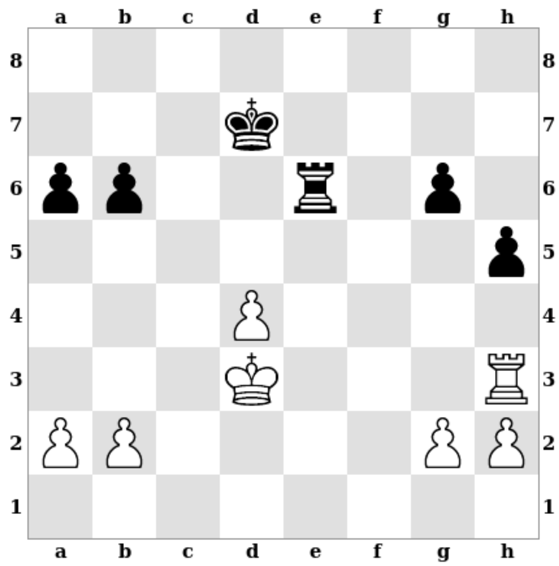
The fact that there are fewer pieces on the board means that it is less likely that a direct attack on one of the kings will be successful. This has two further consequences for strategy. First, it becomes safe for the kings to come out into the game. Second, direct attacks on the king become less important and the focus shifts toward promoting pawns.

King Activity

During the opening and the middlegame, the highest priority for the king is to stay safe and avoid being checkmated. As the endgame approaches, however, it becomes more and more reasonable for the king to enter the game. It turns out that the king is a useful piece in its own right and can often be used to invade the opponent’s territory, win valuable pawns, and defend one’s own pieces and pawns. Bringing the king out in the endgame is just as important as developing the other pieces in the opening.

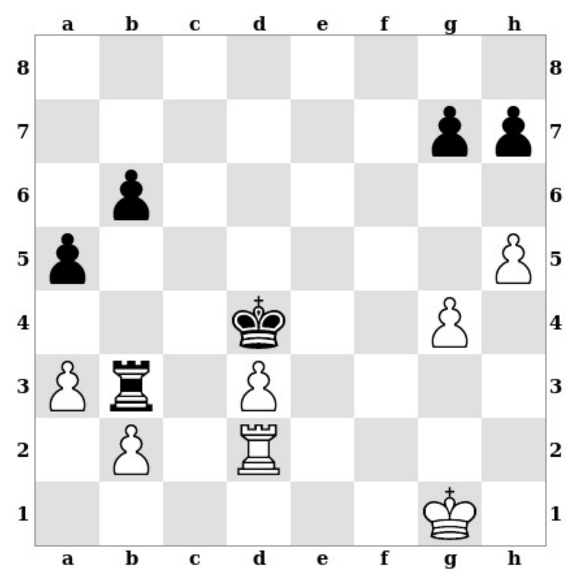


This position occurred in the game Tarrasch vs. Thorold, Manchester, 1890. White has an extra pawn and Black doesn’t currently have any advantages of his own that would be enough to compensate for this material disadvantage. As a result, White has excellent chances to win the game. Tarrasch now brought his king into the game by playing 1.Kf2 g6 2.Rh3 h5 3.Ke3 Rd6 4.d4 Re6+ 5.Kd3, leading to the position in the next diagram.



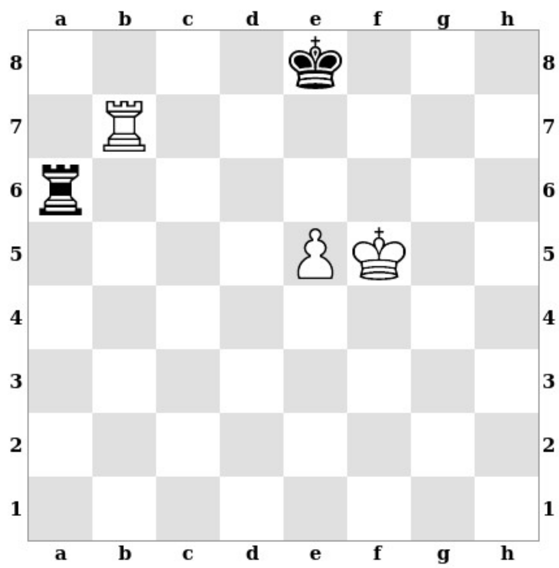
Having activated his king in this way, Tarrasch went on to win the game with his extra pawn.

This diagram, however, shows a position that could have occurred had Tarrasch not activated his king and instead made a series of aimless pawn moves. Black has now activated his own king, while White's king is still uninvolved in the game. Black is still down a pawn for the moment, but he now has a large advantage. There is no way for White to avoid losing his d-pawn to ...Rxd3, after which Black's more active pieces should allow him to win even more material.



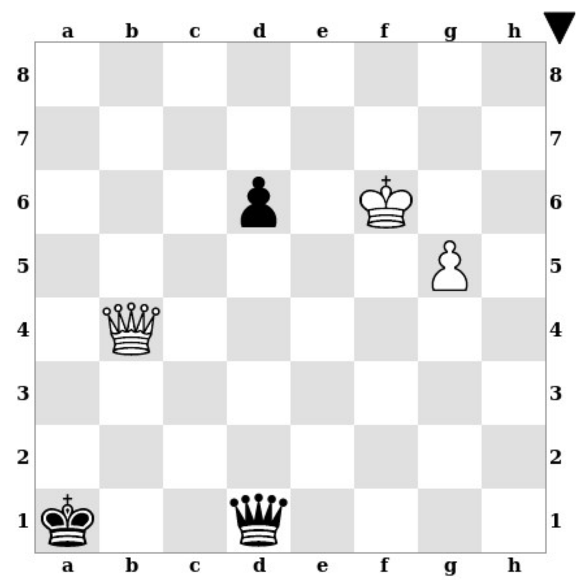
Endgame Theory

Many endgame positions have been given names and studied in much the same way as openings. The difference is that opening theory is often concerned with small advantages, whereas endgame theory is concerned with whether one of the players can win by force or whether the game will end as a draw.



For example, this position is known as Philidor's Position, and is a well-known position in the theory of rook endgames. With correct play, Black can hold a draw no matter how White plays.

Precise evaluations have been calculated for many endgame positions using computers. These results are stored in what are called *endgame tablebases*. At this point, complete results have been computed for all legal position featuring eight or fewer total pieces (counting the kings). For example, this position comes from the game Kasparov vs. The World, Internet, 1999. A tablebase can tell us that White is winning in this position with perfect play, that he can force mate in 82 moves, and also what effect each legal move in the position would have on the outcome. Interfaces for querying tablebase results can be found online in a number of places.



8 – Rules and Exceptions

People often express ideas about chess strategy as generalizations, for example: “knights usually should not be placed on the edge of the board.” The previous chapters of this book contain many generalizations of this kind. People sometimes go further, suggesting that these sorts of generalizations reflect a consistent set of principles that govern the game of chess. In some cases people will even state things as absolute rules, for example: “never put knights on the edge of the board.”

Generalizations are inevitable and quite useful; however, it is important to understand that they are often simplistic and have many exceptions. As strong players have studied chess more deeply, they have become increasingly willing to do things that violate traditional ideas of good strategy. As a result, it seems less and less likely that chess strategy is governed by consistent principles or rules at all.

Given that generalizations about chess strategy are not entirely reliable, it may seem that they have no value. In fact they can be useful, as long as you are careful about how you use them. The best approach is to rely on generalizations only as one part of your thinking about a given position. If you want to move a knight to the edge of the board, for example, you should certainly be aware that this can have negative consequences and you should consider whether or not these are likely in your particular position. But in the end, you should trust your own instincts and calculations over generalizations. The more you play and study chess, the more effectively you will be able to do this and the less you will need to rely on general rules.

As you see less value in generalizations, you may be tempted to ignore any chess idea that is phrased as a generalization or principle. However, there are many great chess books that have a lot of useful things to teach, but which feature an excessive amount of this kind of rule-based thinking. You will also encounter other players and teachers whose ideas about chess have value, but who are overly reliant on principles and rules.

There are two helpful things you can do to deal with these situations. First, rephrase chess ideas in your own words to make them more reasonable. If a book says that “knights do not belong on the edge of the board,” take this to mean that “knights on the edge of the board sometimes become a problem.” If a coach tells you that “you should never move the same piece twice in the opening,” you should understand this to mean that “moving the same piece twice in the opening is sometimes a tempting mistake.”

Second, think about the reasoning behind these claims. Why do knights ever run into trouble on the edge of the board? When does this happen? Why might it be a bad idea to move the same piece repeatedly in the opening? Are there certain times when this is a better or worse idea than others? The very process of considering these questions will help you to stay flexible in your use of these ideas.

You should remember that chess is full of people with opinions about things. Some of those opinions are reasonable, some are simply wrong, and others are entirely a matter of preference or taste. You will learn the most by assuming that people are fallible and by thinking for yourself about everything you are told. As Grandmaster David Bronstein once said, “the essence of chess is thinking about what chess is.”

Appendix - Supplemental Games

Chapter 2 - Material

Game 1
Morphy – Loewenthal
New Orleans, 1850

1.e4 c5 2.f4 e6 3.Nf3 d5 4.exd5 exd5 5.d4 Bg4 6.Be2 Bxf3 7.Bxf3 Nf6 8.O-O Be7 9.Be3 cxd4 10.Bxd4 O-O 11.Nc3 Nc6 12.Bxf6 Bxf6 13.Nxd5 Bxb2 14.Rb1 Bd4+ 15.Kh1 Rb8 16.c3 Bc5 17.f5 Qh4 18.g3 Qg5 19.f6 Ne5 20.fxg7 Rfd8 21.Be4 Qxg7 22.Qh5 Rd6 23.Bxh7+

This begins a series of tricky tactical moves for both sides.

23...Kf8 24.Be4 Rh6 25.Qf5 Qxg3 26.Rb2 Re8 27.Nf6 Re6 28.Rg2 Qxg2+ 29.Bxg2 Rexf6 30.Qxf6 Rxf6 31.Rxf6

The end result is that White is up an exchange (a rook for a minor piece). This two point material advantage should be enough to win.

31...Ng4 32.Rf5 b6 33.Bd5 Nh6 34.Rf6 Kg7 35.Rc6 a5 36.Rc7 Kg6 37.Kg2 f6 38.Kf3 Nf5 39.Be4

White forces a trade of pieces since he has a material advantage.

39...Kg5 40.Bxf5 Kxf5 41.h4 Kg6 42.Rc6 Kh5 43.Kg3 f5 44.Rf6 f4+ 45.Kxf4 Bf2 46.Ke4 Bc5 47.Rf5+ Kxh4 48.Rxc5

White temporarily gives back his material advantage to reach a winning pawn endgame.

48...bxc5 49.Kd5 1-0

Game 2
Botvinnik – Euwe
The Hague, 1948

1.d4 d5 2.c4 e6 3.Nf3 Nf6 4.Nc3 c6 5.e3 Nbd7 6.Bd3 Bb4 7.a3 Ba5 8.Qc2 Qe7 9.Bd2 dxc4 10.Bxc4 e5 11.O-O O-O 12.Rae1 Bc7 13.Ne4 Nxe4 14.Qxe4 a5 15.Ba2 Nf6 16.Qh4 e4 17.Ne5 Bxe5

Black wins a pawn but must give up the bishop pair to do so.

18.dxe5 Qxe5 19.Bc3 Qe7 20.f3

Black now turns down the chance to hold on to his extra pawn with 20...exf3, which he must have decided was too risky.

20...Nd5 21.Qxe7 Nxe7 22.fxe4

White's e-pawns are weak, but his bishop pair and active pieces give him the advantage.

22...b6 23.Rd1 Ng6 24.Rd6 Ba6 25.Rf2 Bb5 26.e5 Ne7 27.e4 c5 28.e6 f6 29.Rxb6 Bc6 30.Rxc6 Nxc6 31.e7+ Rf7 32.Bd5 1-0

Chapter 3 – Pawn Structure

Game 3
Rubinstein – Salwe
Lodz, 1908

1.d4 d5 2.c4 e6 3.Nc3 c5 4.cxd5 exd5 5.Nf3 Nf6 6.g3 Nc6 7.Bg2 cxd4 8.Nxd4 Qb6 9.Nxc6 bxc6

Black's c and d-pawns are now potential weaknesses.

10.O-O Be7 11.Na4

White wants to control the d4 and c5 squares so that Black's weak pawns cannot advance. Notice how many of White's moves increase his control of d4 and c5.

11...Qb5 12.Be3 O-O 13.Rc1 Bg4 14.f3 Be6 15.Bc5 Rfe8 16.Rf2 Nd7 17.Bxe7 Rxe7 18.Qd4 Ree8 19.Bf1 Rec8 20.Nc5 Nxc5 21.Rxc5 Qb6 22.e3 Rc7 23.Rfc2 Rac8 24.b4 a6 25.Ra5 Rb8 26.a3

White has succeeded in immobilizing Black's pawns. Not only are Black's pawns weak, but his pieces have become quite passive and tied down to the task of defending the pawns.

26...Ra7 27.Rxc6 Qxc6 28.Qxa7 Ra8 29.Qc5 Qb7 30.Kf2 g6 31.Qd6 Qc8 32.Rc5 Qb7 33.h4 a5 34.Rc7 Qb8 35.b5 a4 36.b6 Ra5 37.b7 1-0

Game 4
Chekhover – Rudakovsky
Baku, 1945

1.d4 d5 2.c4 e6 3.Nf3 Nf6 4.Bg5 Be7 5.e3 O-O 6.Nc3 Nbd7 7.Qc2 c6 8.Bd3 dxc4 9.Bxc4 Nd5 10.Bxe7 Qxe7 11.O-O b5

This leaves the c6-pawn and the c5-square weak.

12.Be2 a6 13.Ne4

White works to control c5 so that Black's c-pawn cannot advance

13...Bb7 14.Ne5 Rac8 15.Nxd7 Qxd7 16.Nc5 Qc7 17.Rfd1 Rcd8 18.Rac1 Bc8 19.Qe4 Nf6 20.Qh4 Qa5 21.a3 b4 22.a4

White avoids pawn trades and prevents Black from opening lines on the queenside or eliminating any of his weaknesses

22...Nd7 23.b3 Nxc5 24.Rxc5 Qb6 25.Rdc1 Bb7 26.a5 Qa7 27.Bd3 g6 28.Qf6

White now begins looking for possibilities on the kingside as well. With Black's pieces so passive it is hard for him to defend on both sides of the board at once

28...Rd6 29.Qe7 Rfd8 30.h4 R8d7 31.Qf6 Qa8 32.Be4 Qe8 33.h5 Rd8 34.Bxc6 Bxc6 35.h6 Kf8 36.Rxc6 Rxc6 37.Rxc6 Rd7 38.Rc8 Qxc8 39.Qh8+ 1-0

Game 5
Bernstein – Mieses
Coburg, 1904

1.e4 c5 2.Nc3 e6 3.Nf3 Nc6 4.d4 cxd4 5.Nxd4 Nf6 6.Nxc6 bxc6 7.e5 Nd5 8.Ne4

All of Black's central pawns are now on light squares, leaving his dark squares potentially weak.

8...f5 9.exf6 Nxf6 10.Nd6+ Bxd6 11.Qxd6

By trading off the dark-square bishop Black has lost one of the defenders of his dark squares.

11...Ne4 12.Qd4 Nf6 13.Qd6 Ne4 14.Qb4 d5 15.Bd3 Qd6 16.Qxd6 Nxd6 17.f4

White is trying to keep the dark squares under control and prevent Black's pawns from advancing.

17...a5 18.Be3 Ba6 19.Kd2 Nc4+

Allowing White to trade his light-square bishop for the knight further reduces Black's control of the dark squares.

20.Bxc4 Bxc4 21.a4 Kd7 22.b3 Ba6 23.Bb6 Bc8 24.Ke3 Ra6 25.Bc5 Kc7 26.Kd4 Bd7 27.Rhe1 h5 28.Re5 g6 29.Rg5 Rg8 30.Ke5 Bc8 31.Re1 Ra8 32.Kf6

White continues using the dark squares to invade Black's position.

32...Bd7 33.g3 Rae8 34.Ree5 Rh8 35.Rxg6 Rh7 36.Rg7 Reh8 37.Rxh7 Rxh7 38.Kg6 Rh8 39.Kg7 Rd8 40.Rxh5 Be8 41.Rh7 Rd7+ 42.Kh6 Rxh7+ 43.Kxh7 Bh5 44.h4 Bd1 45.c3 Bxb3 46.g4 Kd7 47.g5 e5 48.f5 Bxa4 49.f6 1-0

Game 6
Smyslov – Karpov
Leningrad, 1971

1.c4 c5 2.Nf3 Nf6 3.Nc3 d5 4.cxd5 Nxd5 5.e3 e6 6.d4 cxd4 7.exd4

White now has an isolated pawn on d4 which is a potential long-term weakness. However, in the short-term this pawn gives White more space and some useful open lines for his pieces. Strong players accept weak pawns regularly in return for active play.

7...Be7 8.Bd3 O-O 9.O-O Nc6 10.Re1 Nf6 11.a3 b6 12.Bc2 Bb7 13.Qd3

White isn't threatening Qxh7 yet because of the knight on f6, but Black needs to be alert to the possibility.

13...Rc8 14.Bg5 g6 15.Rad1

White is building up for the move d4-d5.

15...Nd5 16.Bh6 Re8 17.Ba4 a6 18.Nxd5 Qxd5 19.Qe3 Bf6 20.Bb3 Qh5 21.d5

Black has not adequately controlled d5. White's weak isolated pawn suddenly becomes a dangerous passed pawn.

21...Nd8 22.d6 Rc5 23.d7 Re7 24.Qf4 Bg7 25.Qb8 Qxh6 26.Qxd8+ Bf8 27.Re3 Bc6 28.Qxf8+ Qxf8 29.d8=Q 1-0

Game 7
Smyslov – Uhlmann
Mar del Plata, 1966

1.e4 e6 2.d4 d5 3.Nc3 Bb4 4.e5 Ne7 5.a3 Bxc3+ 6.bxc3 c5 7.Nf3 Bd7 8.a4 Qa5 9.Qd2 Nbc6 10.Be2 Rc8 11.dxc5

This ugly move leaves White's pawns potentially quite weak. White is playing this way because he recognizes that these weak pawns come with advantages. He now has two open files (b and d) for his rooks. If Black captures on c5 then White's dark-square bishop may come to life after a3 or Be3.

11...Ng6 12.O-O O-O 13.Qe3 Qc7 14.Nd4 Qxe5 15.Nb5 Qxe3 16.Bxe3 a6 17.Nd6

White has lost a pawn but his pieces are quite active. His knight on d6 is especially strong.

17...Rc7 18.a5 e5 19.Rfb1 Nd8 20.Rd1

White's rook makes good use of the open files created by all his pawn weaknesses.

20...Bc6 21.Bg4 Ne6 22.Rab1 Ne7 23.g3 f5 24.Bh3 g6 25.f4 d4 26.cxd4 Nd5 27.Bf2 exd4 28.Bxd4 Nxd4 29.Rxd4 Re7 30.Kf2 Nc3 31.Re1 Rxe1 32.Kxe1 Ne4 33.Nxe4 Bxe4 34.c3 Rf6 35.Rd8+ Kg7 36.Rd7+ Rf7 37.Rxf7+ Kxf7 38.Kd2 Ke6 39.c4

White's doubled c-pawns help to keep out Black's king.

39...Kd7 40.Ke3 Kc6 41.Kd4 Kd7 42.Ke5 Bf3 43.Kf6

White now abandons his queenside, believing that he will be fast enough on the kingside to win.

43...Kc6 44.Kg7 Kxc5 45.Kxh7 Bh5 46.Bf1 Kb4 47.Bg2 Kxa5 48.Bxb7 Kb6 49.Bc8 a5 50.Bd7 Kc5 51.h3 Bf3 52.Kxg6 Bc6 53.Bxf5 a4 54.Bb1 a3 55.f5 Be4 56.Ba2 Bd3 57.h4 Kd4 58.h5 Ke5 59.g4 Kf4 60.Kg7 1-0

Game 8
Capablanca – Villegas
Buenos Aires, 1914

1.d4 d5 2.Nf3 Nf6 3.e3 c6 4.Bd3 Bg4 5.c4 e6 6.Nbd2 Nbd7 7.O-O Be7 8.Qc2 Bh5 9.b3 Bg6 10.Bb2 Bxd3 11.Qxd3 O-O 12.Rae1 Qc7 13.e4 dxe4 14.Nxe4 Nxe4 15.Rxe4 Bf6 16.Qe3 c5 17.Ne5 cxd4 18.Nxd7 Qxd7 19.Bxd4 Bxd4 20.Rxd4 Qc7 21.Rfd1 Rfd8

White has a majority of pawns on the queenside, while Black has a majority on the kingside. White now begins advancing his queenside majority to try to create a passed pawn.

22.b4 Rxd4 23.Qxd4 b6 24.g3

White takes a moment to give his king a safe square and guard against potential back rank mates.

24...Rc8 25.Rc1 Rd8 26.Qe3 Kf8 27.c5 bxc5 28.Qe4 Rd5 29.bxc5 g6

White has used some clever tactics to turn his queenside majority into a passed pawn.

30.c6 Kg7 31.a4 Rd6

Black blunders, though White was better in any case.

32.Qe5+ Kf8 33.Qxd6+ Qxd6 34.c7 1-0

Game 9
Buckels – Asadzade
Schwäbisch Gmünd, 2019

1.d4 Nf6 2.Nf3 c5 3.d5 e6 4.Nc3 Nxd5 5.Nxd5 exd5 6.Qxd5 Nc6 7.Bg5 Qb6 8.Ne5 Nxe5 9.Qxe5+ Qe6 10.Qc7 Be7 11.Bxe7 Qxe7 12.O-O-O O-O 13.e3 Qf6 14.Rd2 Qc6 15.Qxc6 dxc6

This is not a bad move, but it comes with a downside: White now has a healthy kingside pawn majority while Black's queenside majority is crippled.

16.Bc4 Re8 17.Rhd1 Kf8 18.a4 Be6

This is a serious mistake. Black should not be so eager to trade pieces given the pawn structure.

19.Bxe6 Rxe6 20.Rd8+ Rxd8 21.Rxd8+ Re8 22.Rxe8+ Kxe8

Black's queenside majority cannot easily create a passed pawn while White's kingside majority can. This fact alone makes this king and pawn endgame winning for White right from the beginning.

23.c4 Kd7 24.Kd2 Ke6 25.e4 Ke5 26.Ke3 g5 27.f3 f5 28.exf5 Kxf5 29.a5 b5

White must avoid giving Black a healthy majority on the queenside. For example, it would now be a horrible mistake to play 30.cxb5? cxb5 when White's advantage is gone.

30.b3 a6 31.g3 Ke5 32.f4+ Kf5 33.h3 bxc4 34.bxc4 h6 35.g4+ Kf6 36.Ke4 Ke6 37.h4 gxh4 38.Kf3 Kf7 39.Kg2 Kg7 40.Kh3 Kg6 41.Kxh4 Kf6 42.Kh5 Kg7 43.f5 1-0

Chapter 4 – King Safety

Game 10
Blackburne – Blanchard
London, 1891

1.e4 e5 2.f4 Bc5 3.Nc3 Nc6 4.Nf3 exf4 5.d4 Bb4 6.Bxf4 d5 7.e5 Bxc3+ 8.bxc3 Be6 9.Bd3 h6 10.O-O Nge7 11.Rb1 b6 12.Qd2 O-O

Black's king might appear to be safe here: he has castled and the king has good pawn cover. However, a few things call this into question. First, White has a pawn on e5 giving him a space advantage on the kingside. Second, White's pieces are quite active and many of them are pointing towards Black's king. Finally, Black's pawn on h6 is a good target for sacrifices.

13.Bxh6 gxh6 14.Qxh6 Ng6 15.Ng5 Re8 16.Rxf7 Bxf7 17.Qh7+ Kf8 18.Qxf7# 1-0

Game 11
Zeissl – Walthoffen
Vienna, 1898

1.e4 e5 2.Nf3 Nc6 3.Bb5 f5 4.d4 fxe4 5.Nxe5 Nxe5 6.dxe5 c6 7.Bc4 Qa5+ 8.Nc3 Qxe5 9.O-O d5

White has lost a pawn, but more importantly he has given Black a huge pawn center.

10.Bb3 Nf6 11.Be3 Bd6

Black's goal with this move is to force White to play g3, critically weakening the light squares around White's king.

12.g3 Bg4 13.Qd2 Bf3 14.Bf4 Qh5 15.Nd1 Qh3 16.Ne3 Ng4 17.Rfe1 Qxh2+ 18.Kf1 Qh1# 0-1

Game 12
Greco – Amateur
Rome, 1619

1.e4 e6 2.d4 Nf6 3.Bd3 Nc6 4.Nf3 Be7 5.h4 O-O 6.e5 Nd5

Black's king might look safe enough; he has castled and there are three unmoved pawns in front of the king. However, White has a space advantage on the kingside and a number of pieces pointing toward that side of the board.

7.Bxh7+ Kxh7 8.Ng5+ Bxg5 9.hxg5+ Kg6 10.Qh5+ Kf5 11.Qh7+ g6 12.Qh3+ Ke4 13.Qd3# 1-0

Game 13
Karpov – Gik
Moscow, 1968

1.e4 c5 2.Nf3 d6 3.d4 cxd4 4.Nxd4 Nf6 5.Nc3 g6 6.Be3 Bg7 7.f3 O-O 8.Bc4 Nc6 9.Qd2 Qa5 10.O-O-O

With the kings castled on opposite sides of the board, the players can take greater risks in attacking one another's kings.

10...Bd7 11.h4 Ne5 12.Bb3 Rfc8 13.h5 Nxh5

White's pawn sacrifice has opened the h-file for his rook.

14.Bh6 Bxh6 15.Qxh6 Rxc3

Black makes a sacrifice of his own, giving up the exchange to open White's king position.

16.bxc3 Qxc3 17.Ne2 Qc5 18.g4 Nf6 19.g5 Nh5 20.Rxh5

Another sacrifice, again with the goal of opening up the opposing king.

20...gxh5 21.Rh1 Qe3+ 22.Kb1 Qxf3 23.Rxh5 e6 24.g6 Nxg6 25.Qxh7+ Kf8 26.Rf5 Qxb3+ 27.axb3 exf5 28.Nf4 Rd8 29.Qh6+ Ke8 30.Nxg6 fxg6 31.Qxg6+ Ke7 32.Qg5+ Ke8 33.exf5

The attacks are over. White's material advantage is small but his queen is much more active than Black's pieces and his passed pawn is dangerous.

33...Rc8 34.Qg8+ Ke7 35.Qg7+ Kd8 36.f6 1-0

Game 14
Aloni – Bronstein
Moscow, 1956

1.d4 Nf6 2.c4 g6 3.Nc3 Bg7 4.e4 d6 5.f3 e5 6.Nge2 O-O 7.Be3 Nbd7 8.Qd2 a6 9.g4 exd4 10.Nxd4 c5 11.Nc2 Ne5 12.Be2 Be6 13.Na3 Nfd7 14.O-O-O

When the players castle on opposite sides the game often becomes a race to see whose attack can land first.

14...b5

Black doesn't care about losing the b-pawn; he wants open lines toward White's king.

15.cxb5 axb5 16.Ncxb5 c4 17.Qxd6 Qa5 18.Bd4 Rfc8 19.Bc3 Nd3+ 20.Bxd3 Bxc3 21.Nxc3 cxd3 22.Rxd3

Black is down three pawns but White's king is in serious danger.

22...Ne5 23.Re3 Rd8 24.Qe7 Rd7 25.Qf6 Qc5 26.Nc2 Nd3+ 27.Rxd3 Rxd3 28.a3 Qf2 29.Re1 Rd2 0-1

Chapter 5 – Piece Activity

Game 15
Smyslov – Rudakovsky
Moscow, 1945

1.e4 c5 2.Nf3 e6 3.d4 cxd4 4.Nxd4 Nf6 5.Nc3 d6 6.Be2 Be7 7.O-O O-O 8.Be3 Nc6 9.f4 Qc7 10.Qe1 Nxd4 11.Bxd4 e5 12.Be3 Be6 13.f5 Bc4

In giving up his light-square bishop, Black is starting to lose some influence over d5.

14.Bxc4 Qxc4

White would now like to establish his knight on d5. However, the immediate 15.Nd5 Nxd5 doesn't do much, so White first trades his bishop for Black's knight.

15.Bg5 Rfe8 16.Bxf6 Bxf6 17.Nd5

White's knight outpost now gives him a lasting advantage.

17...Bd8 18.c3 b5 19.b3 Qc5+ 20.Kh1 Rc8 21.Rf3 Kh8 22.f6 gxf6 23.Qh4 Rg8 24.Nxf6 Rg7 25.Rg3 Bxf6 26.Qxf6 Rg8 27.Rd1 d5 28.Rxg7 1-0

Game 16
Eliskases – Flohr
Semmering, 1937

1.d4 Nf6 2.c4 g6 3.Nc3 d5 4.Bf4 Bg7 5.e3 O-O 6.Nf3 c5 7.cxd5 Nxd5 8.Be5 Nxc3 9.bxc3 cxd4 10.Bxg7 Kxg7 11.cxd4 Qa5+ 12.Qd2 Nc6 13.Be2 Rd8 14.Qxa5 Nxa5 15.O-O Be6 16.e4 Bg4 17.Rfd1 e6 18.Kf1 Bxf3 19.Bxf3 Rac8 20.Rd2 e5 21.d5

White's d-pawn is now a protected passed pawn and could potentially be dangerous.

21...Nc4 22.Re2 Nd6

This is an excellent square for Black's knight. From d6 it blockades White's d-pawn while still attacking a number of useful squares, especially e4 and b7. Knights make good blockaders for exactly this reason: their ability to jump means that they usually do not become passive when blockading.

23.Rb1 Rc4 24.g3 Rdc8 25.Bg2 Rc1+ 26.Rxc1 Rxc1+ 27.Re1 Rxe1+ 28.Kxe1

The remainder of the game shows the superiority of Black's knight to White's bishop.

28...f5 29.f3 fxe4 30.fxe4 b5 31.Kd2 a5 32.Kd3 Kf6 33.Bf3 Ke7 34.h4 h6 35.Bd1 Kd8 36.a4 bxa4 37.Bxa4 Kc7 38.Bc2 Kb6 39.Kc3 Kb5 40.Kb3 Kc5 41.Ka4 Nc4 42.Bb3 Nd2 43.Bc2 Nf1 44.Kxa5 Nxg3 45.Ka4 Nh5 46.Kb3 Kd4 47.Kb4 Nf6 48.d6 g5 49.hxg5 hxg5 50.Kb5 g4 51.Bd1 g3 52.Bf3 Ke3 53.Bh1 Kf2 54.Kc6 g2 55.Bxg2 Kxg2 56.d7 Nxd7 57.Kxd7 Kf3 0-1

Game 17
Burn – Alekhine
Carlsbad, 1911

1.e4 e6 2.d4 d5 3.Nc3 Nf6 4.Bg5 Be7 5.e5 Ne4 6.Bxe7 Qxe7 7.Bd3 Nxc3 8.bxc3

Black's light-square bishop is blocked in by his own pawns and could be a serious long-term problem.

8...c5 9.Nf3 Nc6 10.O-O c4

Black drives White's bishop back and gains some queenside space, but he is also putting yet another pawn on a light square which certainly won't help his bad bishop.

11.Be2 Bd7 12.Qd2 b5 13.Ne1 a5 14.a3 O-O 15.f4 b4 16.axb4 axb4 17.Rxa8 Rxa8 18.cxb4 Qxb4 19.c3 Qb3 20.Bd1 Ra2 21.Qc1 Qb6 22.Rf2 Qa7 23.Rxa2 Qxa2 24.Nc2 h6 25.Qa1 Qxa1 26.Nxa1 Na7 27.Kf2 Bc6 28.Ke3 Nb5 29.Kd2 Kf8 30.Nc2 Ke7 31.Ne3 f5 32.Bf3 Kd7 33.g4 fxg4 34.Bxg4 g6 35.Bd1 Ke7 36.Ng4 h5 37.Ne3 Kf7 38.Ng2 Kg7 39.Nh4 Be8 40.Nf3 Kf7 41.Kc2 Bd7 42.Kb2 Na7 43.Ka3 Nc6 44.Ba4 Ke7 45.Nh4 Kf7 46.Bxc6

White trades his bishop for Black's knight, leaving Black's bad bishop as his only remaining piece.

46...Bxc6

It may not be obvious why Black's bishop is a problem for him. It might even seem that the bishop will be useful since it can guard Black's pawns. However, there are a number of problems for the bishop. First, it can't attack any of White's pawns since they are on dark squares, and thus can't help create any counterplay. Second, the bishop is unable to control the dark squares in Black's territory, meaning that White's king and knight can use those squares unchallenged. Finally, because the bishop is blocked by its own pawns it will eventually start running out of good squares.

47.Kb4 Be8 48.Nf3 Ke7 49.Ng5 Bc6 50.Ka3

White takes his time and tries some different piece placements before commiting to anything. He may not need to do this but Black can't do much in the meantime. The e6 pawn is a long-term weakness.

50...Bd7 51.Kb2 Ba4 52.Kc1 Bb3 53.Nf3 Ba4 54.Nh4 Kf7 55.Ng2 Bd7 56.h4 Be8 57.Kb2 Ba4 58.Ne3 Ke7 59.Ka3 Bc6 60.Kb4 Kd7 61.Ka5 Kc7 62.Nc2 Kb7 63.Nb4 Bd7 64.Na6

An important point is that Black will be losing in almost any king and pawn ending. For example, 64...Kc6 65.Nb8+ followed by Nxd7 and Kb6 wins for White.

64...Be8 65.Nc5+ Kc6 66.Nxe6 Bd7 67.Ng5 Bf5 68.Kb4 Bg4 69.Ka3 Kd7 70.Nf7 Be6 71.Nd6 Kc6 72.Kb2 Bg4 73.Kc2 Kd7 74.Kd2 Ke6 75.Ke3 Bh3 76.f5+ gxf5 77.Kf4 Bg4 78.Kg5 Bh3 79.Ne8 Kf7 80.Nf6 f4 81.Kxf4 Be6 82.Kg5 1-0

Game 18
Petrosian – Zeinalli
Leningrad, 1946

1.d4 d5 2.Nf3 Nf6 3.c4 e6 4.g3 dxc4 5.Bg2 Be7 6.Nbd2 Nbd7 7.O-O O-O 8.Nxc4 Nb6 9.Nfe5 c6 10.Qb3 Nxc4 11.Qxc4 Bd7 12.b3 Qb6 13.Bb2 Rac8 14.Rfd1 Rfd8 15.Rac1 Be8 16.Nd3 a5 17.Nc5 Nd7 18.Ba3 Nxc5 19.Bxc5 Bxc5 20.Qxc5 Qxc5 21.dxc5

The key feature of this position is the difference between the bishops. White's bishop occupies a long open diagonal pointing into Black's position while Black's bishop is blocked in by its own pawns.

21...Kf8 22.e4 Ke7 23.e5 Rxd1+ 24.Rxd1 Bd7 25.Rd4 f6 26.Ra4 Ra8 27.f4 f5 28.b4 Kd8 29.Bf1 Kc7 30.Rxa5 Rxa5 31.bxa5 Bc8 32.Kf2 g6 33.h3 h5 34.Ke3 Kd8 35.Kd4 Kc7 36.Bc4 Kb8 37.Kc3 Ka7 38.Kb4 Bd7 39.a6

White temporarily sacrifices a pawn to bring his king in. Black can't hold the pawn in the long run.

39...bxa6 40.Ka5 Bc8 41.a3

Zugzwang. Black must already give the pawn back.

41...Bd7 42.Bxa6 Be8 43.Bc8 Bf7 44.Bd7 Kb7 45.h4 Kc7 46.Ka6

Black's bishop is so bad that White can even afford to sacrifice his bishop to get his king further into Black's position.

46...Kxd7 47.Kb7 Be8 48.a4 Kd8 49.a5 Bd7 50.a6 Bc8+ 51.Kb6 Bxa6 52.Kxa6 1-0

Game 19
Smyslov – Tal
Moscow, 1964

1.c4 g6 2.Nc3 Bg7 3.g3 c5 4.Bg2 Nc6 5.b3 e6 6.Bb2 Nge7 7.Na4 Bxb2 8.Nxb2 O-O 9.e3 d5 10.Nf3 Nf5 11.O-O b6 12.Na4 Bb7 13.cxd5 exd5 14.d3 Qf6 15.Qd2 Rad8 16.Rfd1 Rfe8

17.Rab1 Nd6 18.Ne1 d4 19.e4 Qe7 20.Nc2 f5 21.exf5 Ne5 22.f4 Nf3+ 23.Bxf3 Bxf3 24.Re1 Qe2 25.Rxe2 Rxe2 26.Qxe2 Bxe2 27.Nb2 gxf5 28.Re1 Bh5 29.Nc4 Nxc4 30.bxc4 Re8 31.Kf2 Rxe1 32.Kxe1

The game has reached a minor piece endgame. Black's bishop is in good shape: it can move freely and White's queenside pawns give it some targets. A key question will be whether or not White can find a way to activate his knight. At the moment it has no obvious way into the game (Na3 can be met by ...a6).

32...Kf8 33.Kd2 Ke7 34.Ne1 a6 35.a4

White was worried about Black playing ...b5 but this puts another pawn on a light square, giving Black's bishop another target.

35...a5 36.Kc2 Be8 37.Kb3 Bc6

White has defended his queenside pawns, but his knight remains very passive.

38.Ka3 Kf6 39.Kb3 Kg6 40.Ka3 Kh5 41.h3 Kg6 42.Kb3 Kg7 43.Ka3 Kf6 44.Kb3 Be8 45.Ng2 Bh5 46.Kc2 Be2 47.Ne1 Bf1 48.Nf3 Bxh3 49.Ng5 Bg2 50.Nxh7+ Kg7 51.Ng5 Kg6 52.Kd2 Bc6 53.Kc1 Bg2 54.Kd2 Kh5 55.Ne6 Bc6 56.Nc7 Kg4 57.Nd5 Kxg3 58.Ne7 Bd7 59.Nd5 Bxa4 60.Nxb6 Be8 61.Nd5 Kf3 62.Nc7 Bc6 63.Ne6 a4 64.Nxc5 a3 65.Nb3 a2 66.Kc1 Kxf4 67.Kb2 Ke3 68.Na5 Be8 69.c5 f4 70.c6 Bxc6 71.Nxc6 f3 72.Ne5 f2 0-1

Game 20
Alekhine – Yates
London, 1922

1.d4 Nf6 2.c4 e6 3.Nf3 d5 4.Nc3 Be7 5.Bg5 O-O 6.e3 Nbd7 7.Rc1 c6 8.Qc2 Re8 9.Bd3 dxc4 10.Bxc4 Nd5 11.Ne4 f5 12.Bxe7 Qxe7 13.Ned2 b5 14.Bxd5 cxd5 15.O-O a5 16.Nb3 a4 17.Nc5 Nxc5 18.Qxc5 Qxc5 19.Rxc5 b4 20.Rfc1

Rooks do well on open files. White's rooks currently dominate the only open file on the board.

20...Ba6 21.Ne5 Reb8 22.f3 b3 23.a3

White doesn't want to give the Black rooks an open file of their own.

23...h6 24.Kf2 Kh7 25.h4 Rf8 26.Kg3 Rfb8 27.Rc7 Bb5 28.R1c5 Ba6 29.R5c6 Re8 30.Kf4 Kg8 31.h5 Bf1 32.g3 Ba6 33.Rf7 Kh7 34.Rcc7

Rooks can also be quite powerful on the seventh rank, especially if they can both team up there. White now threatens 35.Rxg7+ Kh8 36.Ng6#.

34...Rg8 35.Nd7 Kh8 36.Nf6 Rgf8 37.Rxg7 Rxf6 38.Ke5 1-0

Game 21
Charousek – Pillsbury
Budapest, 1896

1.e4 e5 2.Nc3 Nf6 3.f4 d5 4.d3 d4 5.Nb1 Nc6 6.Nf3 Bg4 7.Be2 Bxf3 8.Bxf3 Bd6 9.fxe5 Nxe5 10.O-O Qd7 11.Bg5 Nfg4 12.Bxg4 Nxg4 13.h3 Ne5 14.Nd2 f6 15.Bf4 Ng6 16.Bxd6 Qxd6 17.Qg4 O-O 18.Nc4 Qe7 19.Rf5 c6 20.h4 Qe6 21.h5 Ne7 22.Rf4 Qxg4 23.Rxg4 Rad8 24.Rf4 h6 25.Re1 Nc8 26.Kh2 Rfe8 27.g4 b6 28.Kg3 Nd6 29.Nxd6 Rxd6 30.Rf5 Ree6 31.Kf4 Re5 32.c3 Rxf5+ 33.gxf5

The game has reached a rook ending. Black should play 33...dxc3 here when he is at least equal. For some reason he chooses not to do this.

33...Rd8 34.c4

Now that White has managed to play c4 he enjoys a few advantages in the position: more space, an active king, and the open g-file, which leads to a backward black pawn on g7. However, Black will be able to defend g7 well enough if that is his only weakness. To use his advantage, White will need to open more files for his rook.

34...Kf7 35.b4 Ke8 36.a4 Rd7 37.Ra1

White prepares to play the pawn break a4-a5 in order to open the a-file for his rook.

37...Kd8 38.a5 Kc7 39.axb6+ Kxb6 40.Ra5 Re7 41.Kf3 Re5 42.c5+ Kb7 43.Ke2 Re8 44.Kd2 Rd8 45.Kc2 a6 46.Kb3 Rd7 47.Kc4 Rd8 48.Ra1 Rd7 49.Re1

White prepares another pawn break to get his rook into Black's position.

49...Kc7 50.e5 fxe5 51.Rxe5 Kb7 52.Re4 Rd5 53.Re7+ Kb8 54.Rxg7 Rxf5 55.Rg6 Rxh5 56.Rxc6 Kb7 57.Rb6+ Ka7 58.Kxd4 Rh1 59.Kd5 h5 60.d4 h4 61.Rh6 Rb1 62.Rh7+ Kb8 63.Kc6 Rxb4 64.d5 Rg4 65.Rh8+ Ka7 66.d6 1-0

Game 22
Capablanca – Tartakower
New York, 1924

1.d4 f5 2.Nf3 e6 3.c4 Nf6 4.Bg5 Be7 5.Nc3 O-O 6.e3 b6 7.Bd3 Bb7 8.O-O Qe8 9.Qe2 Ne4 10.Bxe7 Nxc3 11.bxc3 Qxe7 12.a4 Bxf3 13.Qxf3 Nc6 14.Rfb1 Rae8 15.Qh3 Rf6 16.f4 Na5 17.Qf3 d6 18.Re1 Qd7 19.e4 fxe4 20.Qxe4 g6 21.g3 Kf8 22.Kg2 Rf7 23.h4 d5 24.cxd5 exd5 25.Qxe8+ Qxe8 26.Rxe8+ Kxe8 27.h5 Rf6 28.hxg6 hxg6 29.Rh1 Kf8 30.Rh7

White's rook takes over the seventh rank. From this position it attacks Black's seventh-rank pawns and restricts Black's king to the eighth rank.

30...Rc6 31.g4 Nc4 32.g5 Ne3+ 33.Kf3 Nf5 34.Bxf5 gxf5 35.Kg3

White sacrifices two pawns to get his king to the powerful f6-square.

35...Rxc3+ 36.Kh4 Rf3 37.g6 Rxf4+ 38.Kg5 Re4 39.Kf6

White is currently down material, but his king and rook are on excellent squares and his passed g-pawn is dangerous.

39...Kg8 40.Rg7+ Kh8 41.Rxc7 Re8 42.Kxf5 Re4 43.Kf6 Rf4+ 44.Ke5 Rg4 45.g7+

If Black takes on g7 then White trades rooks and wins the pawn ending.

45...Kg8 46.Rxa7 Rg1 47.Kxd5 Rc1 48.Kd6 Rc2 49.d5 Rc1 50.Rc7 Ra1 51.Kc6 Rxa4 52.d6 1-0

Game 23
Capablanca – Treybal
Carlsbad, 1929

1.d4 d5 2.c4 c6 3.Nf3 e6 4.Bg5 Be7 5.Bxe7 Qxe7 6.Nbd2 f5 7.e3 Nd7 8.Bd3 Nh6 9.O-O O-O 10.Qc2 g6 11.Rab1 Nf6 12.Ne5 Nf7 13.f4 Bd7 14.Ndf3 Rfd8 15.b4 Be8 16.Rfc1 a6 17.Qf2 Nxe5 18.Nxe5 Nd7 19.Nf3 Rdc8 20.c5 Nf6 21.a4 Ng4 22.Qe1 Nh6 23.h3 Nf7 24.g4 Bd7 25.Rc2 Kh8 26.Rg2 Rg8 27.g5

White now has space advantages on both the queenside and kingside. In order to use these advantages, he will eventually need to find a way to break through into Black's position. The tricky point is figuring out when and where to do this.

27...Qd8 28.h4 Kg7 29.h5 Rh8 30.Rh2 Qc7 31.Qc3

White now reorganizes his pieces a bit. Black has so little active play that White can afford to take his time in doing this. Notice that White does not immediately resolve the tension between the h5 and g6 pawns. He leaves open both the possibility of playing hxg6 to open the h-file and playing h5-h6 to close down the kingside.

31...Qd8 32.Kf2 Qc7 33.Rbh1 Rag8 34.Qa1 Rb8 35.Qa3 Rbg8 36.b5

White finally begins opening some files to enter Black's position.

36...axb5 37.h6+

Only once Black has allowed the a-file to be opened does White shut things down on the kingside.

37...Kf8 38.axb5 Ke7 39.b6 Qb8 40.Ra1 Rc8 41.Qb4 Rhd8 42.Ra7 Kf8 43.Rh1 Be8 44.Rha1 Kg8 45.R1a4 Kf8 46.Qa3 Kg8 47.Kg3

The game now features another period in which White seeks the best piece arrangment before proceeding forward.

47...Bd7 48.Kh4 Kh8 49.Qa1 Kg8 50.Kg3 Kf8 51.Kg2 Be8 52.Nd2 Bd7 53.Nb3 Re8 54.Na5 Nd8

White is now ready to force his way in.

55.Ba6 bxa6 56.Rxd7 Re7 57.Rxd8+ Rxd8 58.Nxc6 1-0

Game 24
Taimanov – Najdorf
Zurich, 1953

1.d4 Nf6 2.c4 g6 3.Nc3 Bg7 4.e4 d6 5.Nf3 O-O 6.Be2 e5 7.O-O Nc6 8.d5 Ne7

The center pawns are now blocked and the players have rival space advantages. White's central pawns point to the queenside where he has a space advantage. Black's pawns point toward the kingside where he has a space advantage. White will aim to play c4-c5, open lines on the queenside, and invade there. Black will aim to play f7-f5, open lines on the kingside, and begin making threats to White's king.

9.Ne1 Nd7 10.Be3 f5 11.f3 f4 12.Bf2 g5 13.Nd3 Nf6 14.c5 Ng6 15.Rc1 Rf7 16.Rc2 Bf8 17.cxd6 cxd6 18.Qd2 g4 19.Rfc1 g3

Black sacrifices a pawn to begin opening lines against White's king.

20.hxg3 fxg3 21.Bxg3 Nh5 22.Bh2 Be7 23.Nb1 Bd7 24.Qe1 Bg5 25.Nd2 Be3+ 26.Kh1 Qg5 27.Bf1 Raf8 28.Rd1 b5 29.a4 a6 30.axb5 axb5 31.Rc7 Rg7 32.Nb3 Nh4

White has succeeded in invading on the queenside, but he doesn't have any real threats. Meanwhile, Black's kingside attack has become unstoppable.

33.Rc2 Bh3 34.Qe2 Nxg2 35.Bxg2 Bxg2+ 36.Qxg2 Qh4 37.Qxg7+ Kxg7 38.Rg2+ Kh8 39.Ne1 Nf4 40.Rg3 Bf2 41.Rg4 Qh3 42.Nd2 h5 43.Rg5 0-1

Game 25
Tal – Fischer
Bled, 1959

1.d4 Nf6 2.c4 g6 3.Nc3 Bg7 4.e4 d6 5.Be2 O-O 6.Nf3 e5 7.d5

The central pawn structure now suggests that White will play on the queenside and Black on the kingside.

7...Nbd7 8.Bg5 h6 9.Bh4 a6 10.Nd2 Qe8 11.O-O Nh7 12.b4 Ng5 13.f3 f5 14.Bf2 Qe7 15.Rc1 Nf6 16.c5 Bd7 17.Qc2 Nh5 18.b5 fxe4 19.Ndxe4 Nxe4 20.fxe4 Nf4 21.c6 Qg5 22.Bf3 bxc6 23.dxc6 Bg4 24.Bxg4 Qxg4 25.Be3 axb5 26.Bxf4 exf4 27.Nxb5

White's queenside play has given him an outside passed a-pawn and a dangerously far-advanced c-pawn. The c-pawn is blocked at the moment, but that could change if White either manages to win the c7-pawn or else sacrifices on d6.} **27...Rf7 28.Qc4 Rc8 29.Rf3** {White now shifts his play to focus on the center and kingside. His queenside advantages remain and limit Black's choice of squares for his pieces.

29...Be5 30.Rcf1 Kg7 31.a4 Ra8 32.Kh1 Qg5 33.g3 Raf8 34.gxf4 Bxf4 35.Nd4

White threatens Ne6+ and there is no good defense.

35...Qh4 36.Rxf4 Rxf4 37.Ne6+ Kh8 38.Qd4+ R8f6 39.Nxf4 Kh7 40.e5 dxe5 41.Qd7+ 1-0

Game 26
Morphy – Loewenthal
London, 1858

1.e4 e5 2.f4 Bc5 3.Nf3 d6 4.c3 Bg4 5.Be2 Bxf3 6.Bxf3 Nc6 7.b4 Bb6 8.b5 Nce7 9.d4 exf4 10.Bxf4 Ng6 11.Be3

White's strong pawn center gives him an advantage.

11...Nf6 12.Nd2 O-O 13.O-O h6 14.a4 c6 15.Qe2 Re8 16.Qd3

So far, White has simply developed his pieces and supported his pawn center.

16...d5

Black wants to challenge White's center, but he isn't prepared for this move yet.

17.e5 Nd7 18.Bh5 Re6 19.a5 Bc7 20.Rxf7

An exchange sacrifice breaks open Black's position.

20...Kxf7 21.Qf5+ Ke7 22.Bxg6 Qg8 23.Bf2

White now threatens Bh4+ and can meet 23...Kd8 with 24.Bf7. Black tries a desperate sacrifice, but he is losing in any case.

23...Nxe5 24.dxe5 Rf8 25.Bc5+ Kd8 26.Bxf8 Rxe5 27.Qf2 Qe6 28.b6 axb6 29.axb6 Qxg6 30.bxc7+ Kxc7 31.Rb1 1-0

Chapter 6 – The Opening

Game 27

Morphy – Duke Karl / Count Isouard

Paris, 1858

1.e4 e5 2.Nf3 d6 3.d4 Bg4 4.dxe5 Bxf3 5.Qxf3 dxe5 6.Bc4

White is developing his pieces and making threats at the same time.

6...Nf6 7.Qb3

This is not a developing move, but it also doesn't lose any time. White threatens Bxf7 and Qxb7 and Black has no good developing move of his own to counter these threats.

7...Qe7

This move blocks in Black's bishop and further slows down his development. Black's motivation is probably that he wants to meet 8.Qxb7 with 8...Qb4+, trading off the queens. This would be fine for White since he wins a pawn, but it would at least allow Black to survive for a while.

8.Nc3

White holds off on playing Qxb7 and continues developing instead.

8...c6 9.Bg5

White now has a significant lead in development: he has four pieces developed compared to Black's two. Additionally, Black can't currently develop his bishop and he isn't ready to castle, whereas White is ready to castle to either side

9...b5 10.Nxb5

With such a big lead in development, White decides to sacrifice a piece to open lines and create threats.

10...cxb5 11.Bxb5+ Nbd7 12.O-O-O Rd8 13.Rxd7

A further sacrifice allows White to get his last remining piece into the game.

13...Rxd7 14.Rd1 Qe6 15.Bxd7+ Nxd7 16.Qb8+ Nxb8 17.Rd8# 1-0

Game 28

Ruger – Gebhard

Dresden, 1915

1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3

White prepares d2-d4 to build a strong pawn center.

4...Nf6 5.d4 exd4 6.cxd4 Bb4+ 7.Nc3

Black should now play ...Nxe4 either immediately or on the next move to eliminate one of White's center pawns.

7...O-O 8.d5 Ne7 9.e5

Black has missed his chance and White's pawn center now gives him a significant advantage.

9...Ne4 10.Qc2 Nxc3 11.bxc3 Bc5 12.Ng5 Ng6 13.h4 h6 14.d6 hxg5 15.hxg5 Re8 16.Qxg6 Rxe5+ 17.Kf1 1-0

Game 29
Tarrasch – Kuerschner
Nuremberg, 1889

1.d4 d5 2.c4 dxc4 3.e3 Bf5 4.Bxc4 e6 5.Qb3 Be4 6.f3 Bc6 7.Ne2 Nf6 8.e4 Be7 9.Nbc3 Qc8 10.d5 exd5 11.exd5 Bd7 12.d6 Bxd6 13.Bxf7+ Kd8

Black's king is now stuck in the center, unable to castle.

14.Bg5 Nc6 15.Ne4 Be7 16.Bxf6 gxf6 17.O-O-O

White castles his own king to safety while bringing a rook into the attack.

17...Ne5 18.Nf4

White now makes good use of the pin on the d7 bishop. The current threat is 19.Ne6#.

18...Qb8 19.Qe6 Rf8 20.Nxf6 Bd6 21.Nxd7 Nxd7 22.Rhe1 1-0

Chapter 7 – The Endgame

Game 30
Tarrasch – Mieses
Goteborg, 1920

1.e4 d5 2.exd5 Qxd5 3.Nc3 Qa5 4.d4 e5 5.Nf3 Bb4 6.Bd2 Bg4 7.Be2 exd4 8.Nxd4 Qe5 9.Ncb5 Bxe2 10.Qxe2 Bxd2+ 11.Kxd2 Qxe2+ 12.Kxe2 Na6 13.Rhe1 O-O-O 14.Nxa7+ Kb8 15.Nac6+ bxc6 16.Nxc6+ Kc8 17.Nxd8 Kxd8 18.Rad1+ Ke8 19.Kd3+

In the endgame it is a good idea to activate the king. White brings his king to the queenside where it can make threats of its own and help his pawns to advance.

19...Ne7 20.Kc4 h5 21.Rd3 Nb8 22.Rde3 Nc6 23.b4 f6 24.f4 Kf7 25.a4 Rb8 26.c3 Rd8 27.Rd3 Rxd3 28.Kxd3 Ke8 29.a5 Kd7 30.a6 Nd5 31.Ra1 Na7 32.g3 c6 33.Ra4 Nb6 34.Ra5 g6 35.c4 Nbc8 36.Ra1 Nd6

White now further improves the position of his king.

37.Kd4 Ndc8 38.Kc5 Kc7 39.Re1 Nb6 40.Re7+ Nd7+ 41.Rxd7+

White recognizes that his active king and his queenside pawns can overpower Black's knight.

41...Kxd7 42.b5 cxb5 43.cxb5 Nc8 44.b6 1-0